GRADUATE STUDY EDITION

Hybrid-Electric Bus System Modeling

A MATLAB and Simulink textbook for selectable hybrid and battery-electric architectures, route studies, supervisory control, and engineering evidence

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Preface

This book explains a complete concept-level bus powertrain study environment with selectable hybrid-electric and battery-electric architectures. It is written for graduate students who already know basic mechanics, electrical power, and MATLAB syntax, but who may be new to model-based design and energy-management architecture.

The project combines a transparent backward-demand energy model, constrained-speed and forward-performance formulations, three editable Simulink representations, modular MATLAB/MAT component data, a programmatic MATLAB app, an optional headless configuration-screening API, and automated verification. Its purpose is educational and architectural: it shows how equations, assumptions, data, software interfaces, and test evidence must remain consistent across a system model.

How to read this book
Chapters 1-3 establish the system and data architecture. Chapters 4-10 derive the physical and supervisory models. Chapter 11 explains the current app workflow. Chapter 12 documents optional script-driven screening outside the app. Chapters 13-15 cover routes and validation; Chapters 16-17 interpret worked studies and responsible extensions.

Learning outcomes

- Derive tractive force and wheel power from a prescribed route speed and grade.
- Separate the traction DC-bus path from the genset-to-standby charging path.
- Explain a deterministic 30% dual-battery role-swap strategy and constant-point genset control.
- Configure Hybrid whole battery sets and BEV half-set increments, explain parallel capability-based power sharing, and distinguish charge-depleting from charge-sustaining assessment.
- Derive the auxiliary-first, active-battery-second, resistor-bank-third regeneration allocation and verify its power balance.
- Distinguish energy accounting, equivalent replenishment, charge sustaining comparison, and true feasibility.
- Audit route provenance, component assumptions, numerical implementation, and validation evidence.
- Design extensions that increase fidelity without hiding uncertainty.

Scope statement

This is a concept energy model. It does not claim certification, manufacturer performance, thermal safety, battery ageing, emissions compliance, or real-time controller readiness. Synthetic component values are used deliberately so that the architecture can be studied without implying supplier data.

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Nomenclature and Conventions

Table F.1. Principal symbols and units

Symbol / term	Meaning	Typical unit
v, a	vehicle speed and longitudinal acceleration	m/s, m/s2
m	calculated curb mass plus user-entered load	kg
F_{trac}	net tractive force required at the tyre contact patch	N
P_{wheel}	wheel mechanical power; positive traction, negative braking	kW
$P_{motor,dc}$	motor-side DC bus power; negative during regeneration	kW
P_{reg}	non-negative regenerative DC power available for allocation	kW
P_{dump}	surplus regenerative power dissipated by the resistor load bank	kW
SOE	stored usable battery energy divided by usable-energy rating	fraction or %
BSFC	brake-specific fuel consumption	g/kWh
η	directional efficiency	fraction
C_{rr}	rolling resistance coefficient	1
ρ	air density	kg/m3
Cd, A	drag coefficient and frontal area	1, m2

Sign convention
Battery power is positive when a pack discharges to the traction DC bus and negative while charging. Wheel power is positive for traction and negative for braking. Genset power is positive into the isolated standby charger; it never offsets traction demand.

1 Project Orientation

The project asks a systems question: for a prescribed bus mission, selected component set, loading state, environment, and supervisory calibration, how much wheel energy, battery energy, diesel fuel, grid-replenishment energy, operating cost, and feasible range result?

1.1 Why multiple simulation formulations?

A backward-facing model starts from the route speed that the vehicle is assumed to achieve. It differentiates speed to obtain acceleration, computes the tractive force required at the wheels, and then propagates power demand backward through the driveline to the DC bus and energy sources. This is efficient for energy sizing and comparison because it avoids solving a driver controller and vehicle-speed tracking loop.

The app also provides a fast constrained formulation and a forward Performance formulation. Constrained mode suppresses requested acceleration using available battery current and energy, motor torque-speed/power, final-drive force, and road load while retaining the backward energy kernel. Performance mode closes the loop around desired speed, component limits, and the longitudinal vehicle plant. These modes expose achieved speed, distance completion, tracking error, limiting cause, depletion, and stall behavior.

Interpretation boundary
Use Backward mode for required-energy screening, Constrained mode for fast feasibility screening, and Performance mode when achieved speed and mission completion are the decision variables. A prescribed-speed result alone is not evidence that the vehicle can track the route.

1.2 Project deliverables

Table 1.1. Primary project artifacts

Artifact	Role in the study
data/HybridBus_ComponentDatabase.xlsx	Selections, common calibration, prices, tyre, final-drive, and governance data
data/routes/*.mat; data/batteries/*.m; data/motors/*.m; data/gensets/*.m	Extensible one-file-per-variant route and component definitions
models/HybridBus_BackwardModel.slx	Editable ordinary-block hybrid backward-demand representation
models/HybridBus_BEVModel.slx	Separate editable BEV Simulink model with scalable parallel battery banks and no genset
models/HybridBus_PerformanceModel.slx	Editable forward-performance plant with route, driver, component-limit, and vehicle subsystems
simulate_hybrid_bus_core.m	Detailed discrete reference kernel used by app runs and headless batch studies
HybridBusApp.m	Explorer app for selecting, simulating, comparing, plotting, and exporting
tests/run_all_hybrid_bus_tests.m	Fifty assertion-based physics, data, control, braking, performance, and ranking scenarios
tests/*.m	Fifty-six MATLAB unit and integration tests, including Hybrid/BEV equivalence and extensibility

1.3 Architecture at a glance

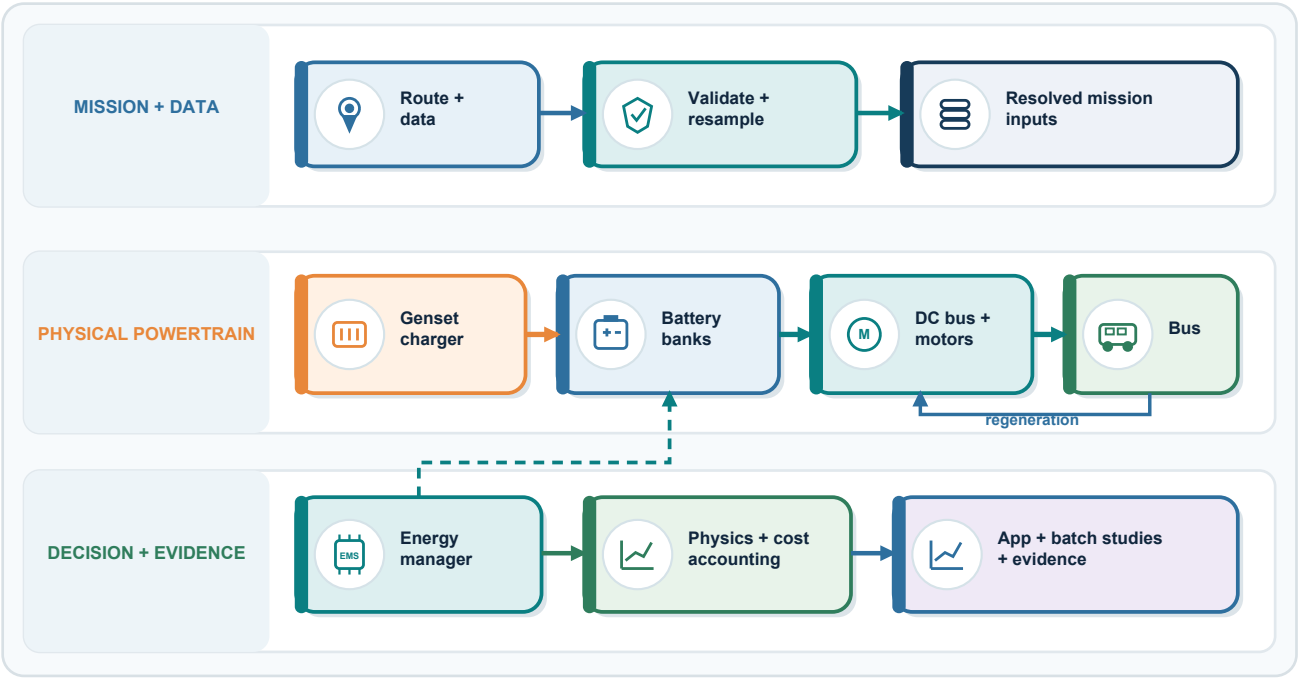


Figure 1.1. End-to-end project architecture from sourced routes to app, batch-study, and evidence outputs

The workbook supplies common calibration and selections, while route and component catalogs are modular MAT and MATLAB files. The loader resolves stable IDs, the validator checks schema and physical ranges, and the input-preparation function resamples the selected route to a 1 s grid. Both the MATLAB kernel and the SLX model consume the same resolved parameter set.

2 Selectable Hybrid and Battery-Electric Architectures

2.1 Energy topology

The architecture contains two independent battery packs, two rear hub motors, a diesel engine-generator set, an auxiliary electrical load, and a resistor load bank. The topology has two deliberately separated electrical paths. Only the active battery connects to the traction DC bus; the genset connects only to the standby battery through a dedicated charger. The genset therefore cannot propel the vehicle or reduce active-battery traction demand. During braking, regenerated electrical power supplies auxiliaries first, charges only the active battery second, and is sent to the resistor load bank if the active battery cannot accept the remainder.

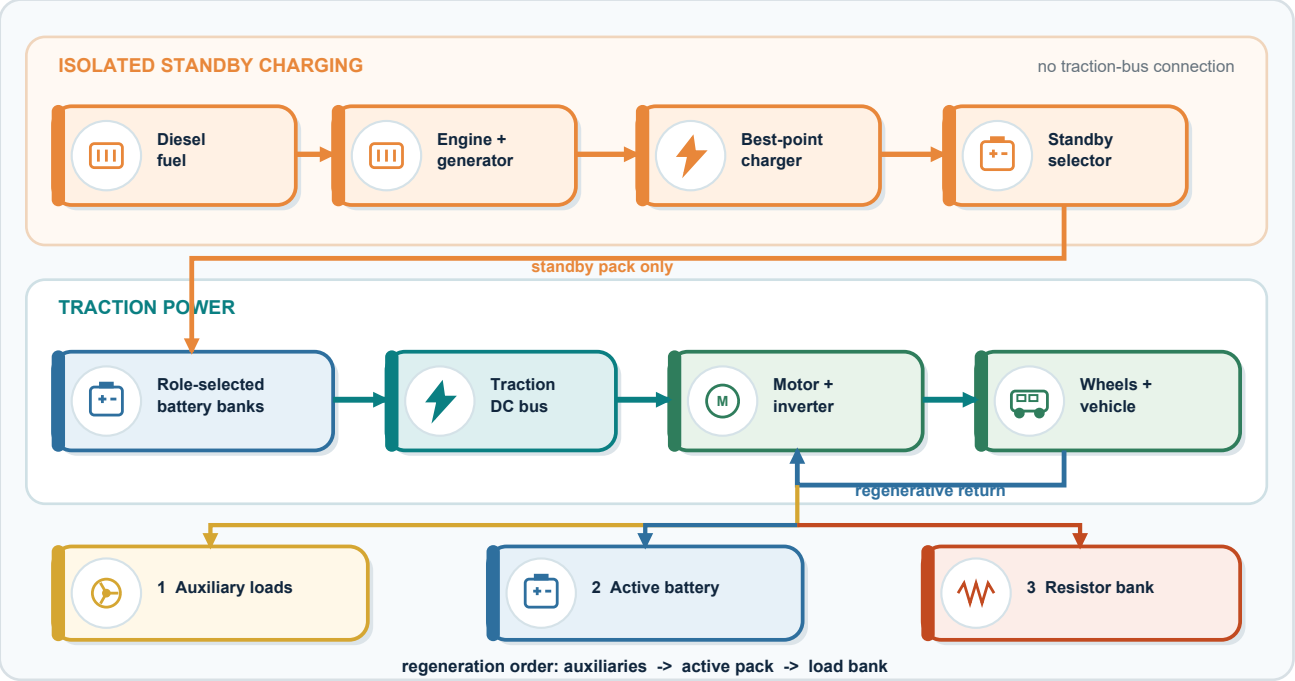


Figure 2.1. Conceptual energy flow, isolated genset path, and regeneration priorities

2.2 Simulink decomposition

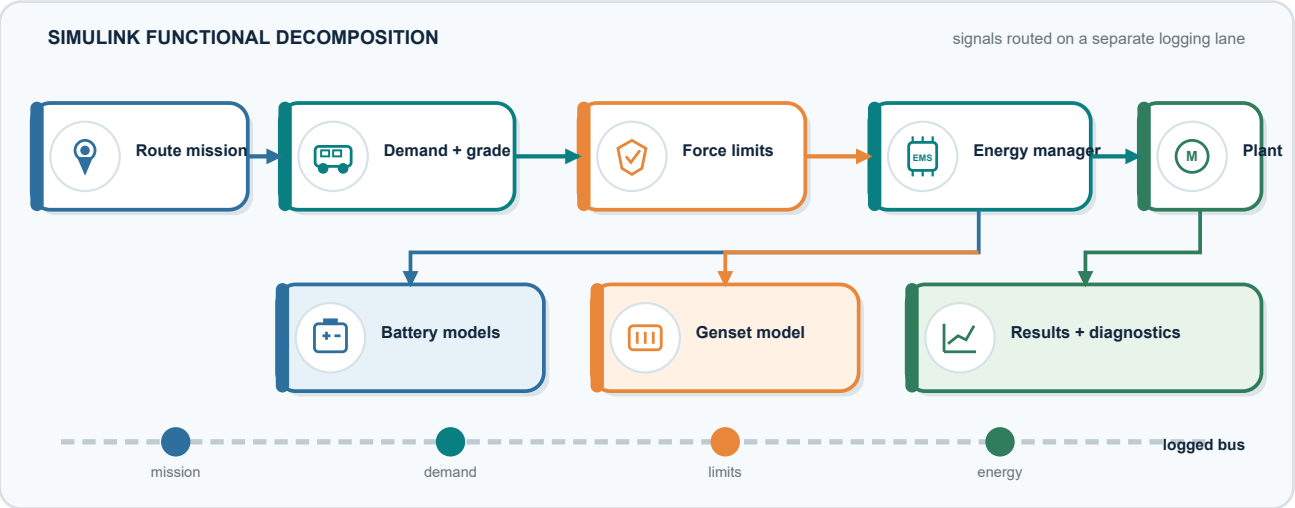


Figure 2.2. Readable Simulink subsystem decomposition and separated logging lane

Table 2.1. Top-level model responsibilities

Subsystem	Responsibility
Route_and_Environment	Publishes route speed, grade, and auxiliary multiplier
Vehicle_Longitudinal_Dynamics	Computes filtered acceleration and required wheel power
Rear_Hub_Motor_Drive	Applies motor torque, power, speed, efficiency, and regeneration limits
Auxiliary_Loads	Calculates base and HVAC electrical demand
Supervisory_Energy_Management	Enforces the 30% role swap, routes fixed genset power only to standby, and applies auxiliary-active-dump regeneration priority
Battery_Pack_1 / Battery_Pack_2	Integrate independent energy states under power and SOE limits
Engine_Genset	Produces constant optimum electrical power when enabled and calculates fuel rate
Energy_and_Cost_Accounting	Includes resistor-bank dissipation in the DC balance and accumulates energy/cost quantities
Output_Logging	Publishes stable workspace signals for inspection

2.3 Reference kernel versus SLX

The ordinary-block SLX is intended for architecture inspection, signal tracing, and interactive simulation. The MATLAB kernel contains the complete deterministic limit handling and supervisory behavior used by app runs and optional batch studies. This separation keeps the teaching model editable while making the numerical reference path easy to test.

2.4 Battery-electric alternative

The Powertrain Architecture switch selects Hybrid or BEV without changing the component database. In BEV mode the diesel fuel, engine, generator, and isolated standby charger are absent. The former Use both batteries checkbox has been removed. A Battery set multiplier now defines installed capacity: 0.5, 1.0, 1.5, and 2.0 BEV sets mean one, two, three, and four connected packs. Odd pack totals assign the additional pack to the Battery 1 selection. All connected packs begin at the same SOE, 85% by default.

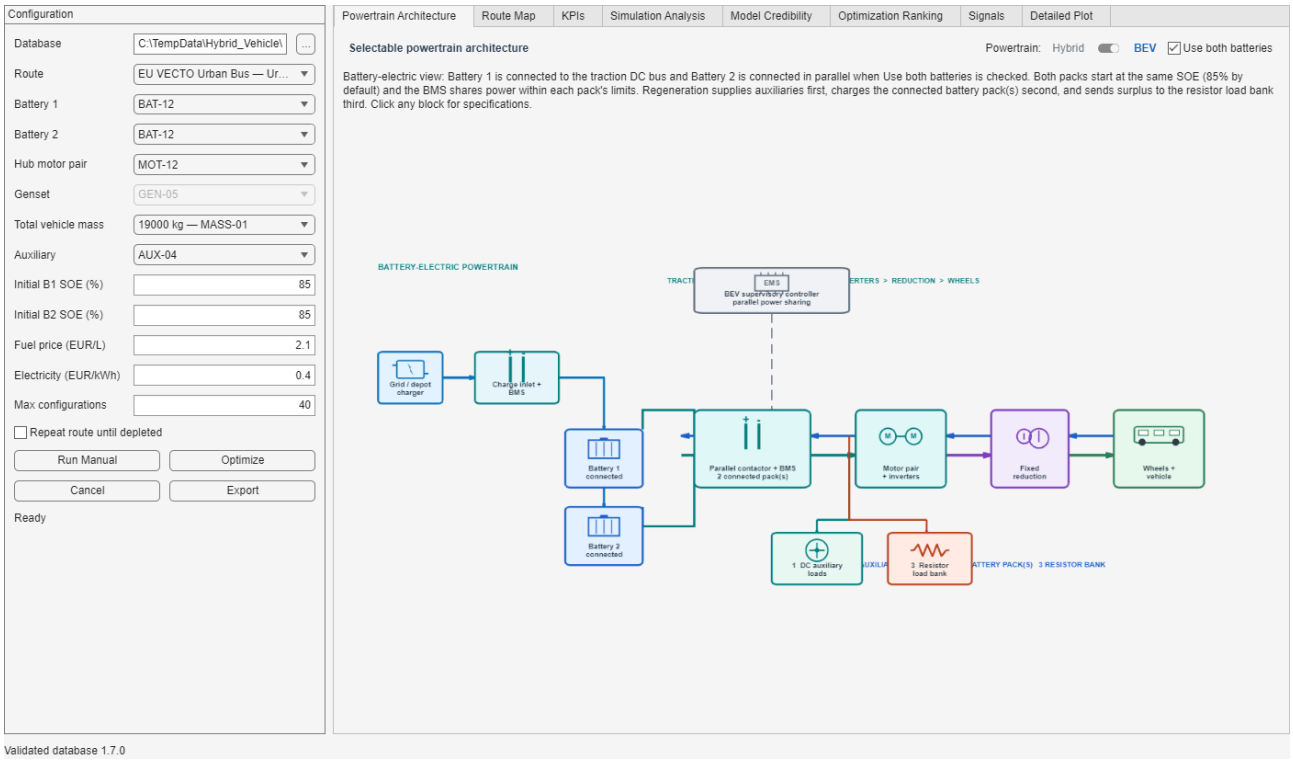


Figure 2.3. Selectable BEV architecture with scalable parallel battery banks, external charging, and regenerative return

For a requested signed DC power Pdc, the controller computes each connected bank's instantaneous charge or discharge capability from its SOE window, scaled pack count, power rating, derating factor, efficiency, and remaining energy headroom. Accepted power is shared in proportion to those capabilities:

$$P_1 = P_{accepted}\, C_1/(C_1+C_2), \quad P_2 = P_{accepted}\, C_2/(C_1+C_2) \tag{2.1}$$

Regeneration retains the same first-principles destination order: auxiliaries first, connected battery pack or packs second, and the resistor load bank third. The BEV kernel sets genset power, genset starts, fuel rate, and fuel use identically to zero. HybridBus_BEVModel.slx is a separate editable fixed-step representation using the same database-resolved route, motor, driveline, auxiliary, battery, vehicle, and economic parameters.

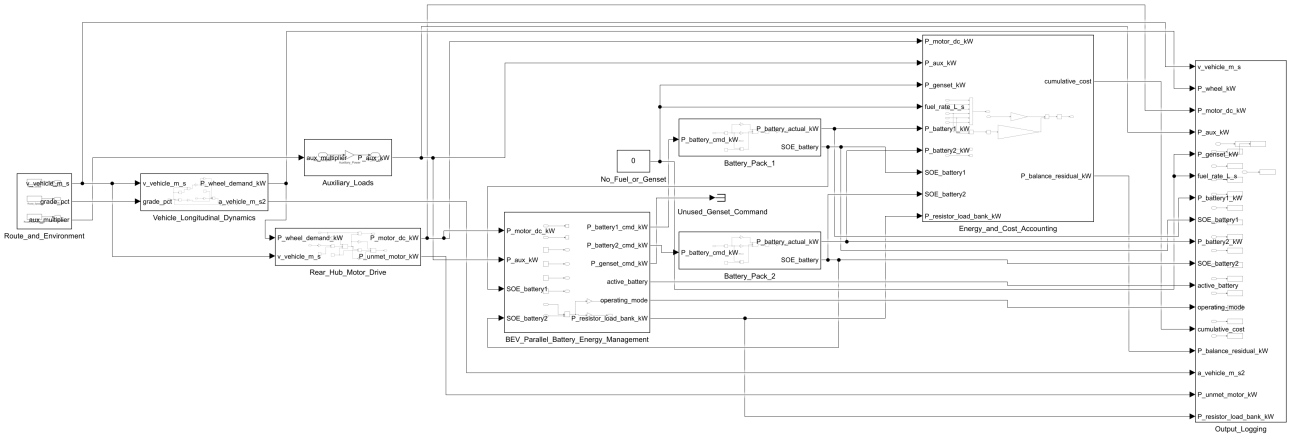


Figure 2.4. Separate HybridBus_BEVModel.slx with parallel battery energy management and explicit zero fuel/genset path

2.5 Battery-set interpretation in Hybrid mode

Hybrid accepts only positive whole-number set multipliers. One set contains one Battery 1 pack in one role bank and one Battery 2 pack in the alternate role bank. A multiplier of two therefore produces two Battery 1 packs in parallel and two Battery 2 packs in parallel. Exactly one bank is active for traction while the equal-sized alternate bank is standby; the 30% role-swap rule transfers those bank roles, not individual packs.

$$N_{B1,Hybrid} = N_{B2,Hybrid} = k, \quad k = 1,2,3,... \tag{2.2}$$

$$N_{packs,BEV} = 2k, \quad k = 0.5,1.0,1.5,... \tag{2.3}$$

3 Data Architecture and Reproducible Inputs

3.1 Workbook plus modular engineering data

The version 1.8.0 workbook retains dashboard selections, common vehicle and control calibration, prices, tyre and final-drive data, and governance sheets. Route histories are stored one route per MAT file. Each battery and motor variant is one MATLAB data function, while each genset MATLAB file contains the matched genset, engine, generator, fuel map, and generator-efficiency map. IDs rather than row numbers form every interface.

Table 3.1. Hybrid project data organization

Data group	Storage	Purpose
Selections and calibration	Workbook Dashboard, Vehicle, Environment, Control	Chosen IDs, SOEs, prices, vehicle constants, and controls
Routes	data/routes/*.mat	Independent time, distance, grade, geometry, elevation, provenance, and license records
Battery variants	data/batteries/*.m	SOE/temperature current, OCV, resistance, energy, mass, and compatibility data
Motor variants	data/motors/*.m	Torque-speed limits, two-dimensional loss maps, mass, and compatibility data
Genset assemblies	data/gensets/*.m	Matched engine, generator, fuel, efficiency, and assembly data
Common hardware	Workbook Tyre and Final Drive	Shared driveline ratings and efficiency
Governance	Workbook definitions/change log + extension guide	Schema, units, provenance, and safe add-on procedure

3.2 Input pipeline

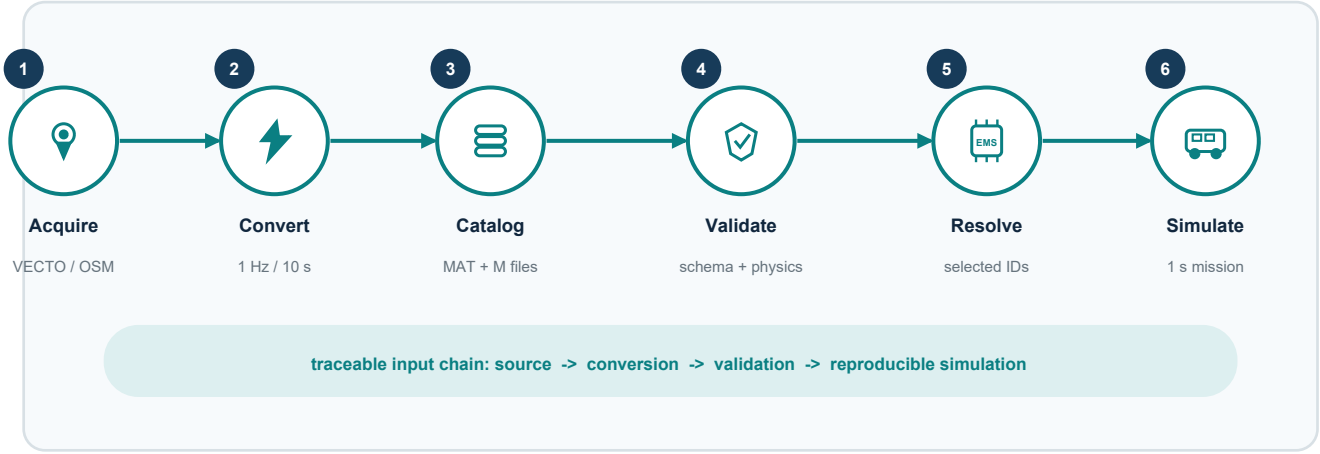


Figure 3.1. Route and component data pipeline

The loader imports workbook calibration, recursively discovers component MATLAB files, and reads every route MAT file. Validation checks schemas, case-insensitive unique IDs, cross-references, lookup dimensions and monotonic breakpoints, physical ranges, route finiteness, monotonic time and distance, and nonzero mission length. Preparation resolves the selected records and resamples speed, grade, elevation, and auxiliary multiplier at the model sample time.

3.3 Default configuration

Table 3.2. Default case resolved from the Dashboard

Input	Selection / value	Unit or interpretation
Route	VECTO-URBAN	stable route ID
Battery 1 / 2	BAT-12 / BAT-12	independent packs
Motor / genset	MOT-12 / GEN-05	two identical hub motors + one genset
Battery set multiplier / load	1.0 / 0.0	sets / tonnes
Calculated curb / total mass	18.764 / 18.764	tonnes
Initial B1 / B2 SOE	85 / 20	%
Fuel / electricity price	2.10 / 0.40	EUR/L / EUR/kWh

4 Longitudinal Vehicle Dynamics

4.1 Speed, acceleration, and grade

The route speed is prescribed. A first-order discrete filter is applied to the raw finite-difference acceleration to reduce numerical spikes without introducing a continuous state.

$$a_{raw,k} = (v_k - v_{k-1}) / \Delta t_{k-1} \quad (4.1)$$

$$a_k = a_{k-1} + [\Delta t / (\tau + \Delta t)] (a_{raw,k} - a_{k-1}) \quad (4.2)$$

Road grade is stored as percent. The model converts it to an angle using $\theta = \text{atan}(\text{grade}/100)$. Headwind is added to vehicle speed and clipped so aerodynamic relative speed cannot become negative.

4.2 Road-load forces

$$F_{inertia} = m\, a \quad (4.3)$$

$$F_{roll} = m\, g\, C_{rr} \cos(\theta) \quad (4.4)$$

$$F_{grade} = m\, g \sin(\theta) \quad (4.5)$$

$$F_{aero} = 0.5\, \rho\, C_d\, A\, v_{air}^2 \quad (4.6)$$

$$F_{tractive} = F_{inertia} + F_{roll} + F_{grade} + F_{aero} \quad (4.7)$$

$$P_{wheel,demand} = F_{tractive}\, v / 1000 \quad (4.8)$$

Mass definition
The fixed mass selector has been retired. Calculated curb mass equals a 15,000 kg base vehicle plus every installed battery pack and, in Hybrid mode only, the selected complete genset assembly. Total vehicle mass equals calculated curb mass plus the user-entered passenger, luggage, and cargo load. Motors and driveline are treated as part of the 15-tonne base and are not added again.

$$m_{curb} = 15000 + N_{B1} m_{B1} + N_{B2} m_{B2} + I_{Hybrid} m_{genset} \quad (4.9)$$

$$m_{total} = m_{curb} + 1000\, L_{tonnes} \quad (4.10)$$

4.3 Force scaling intuition

Inertia, rolling resistance, and grade forces scale linearly with total mass. Aerodynamic force does not. Therefore mass strongly changes stop-and-go and climbing demand, while high-speed cruise also depends heavily on frontal area and drag coefficient.

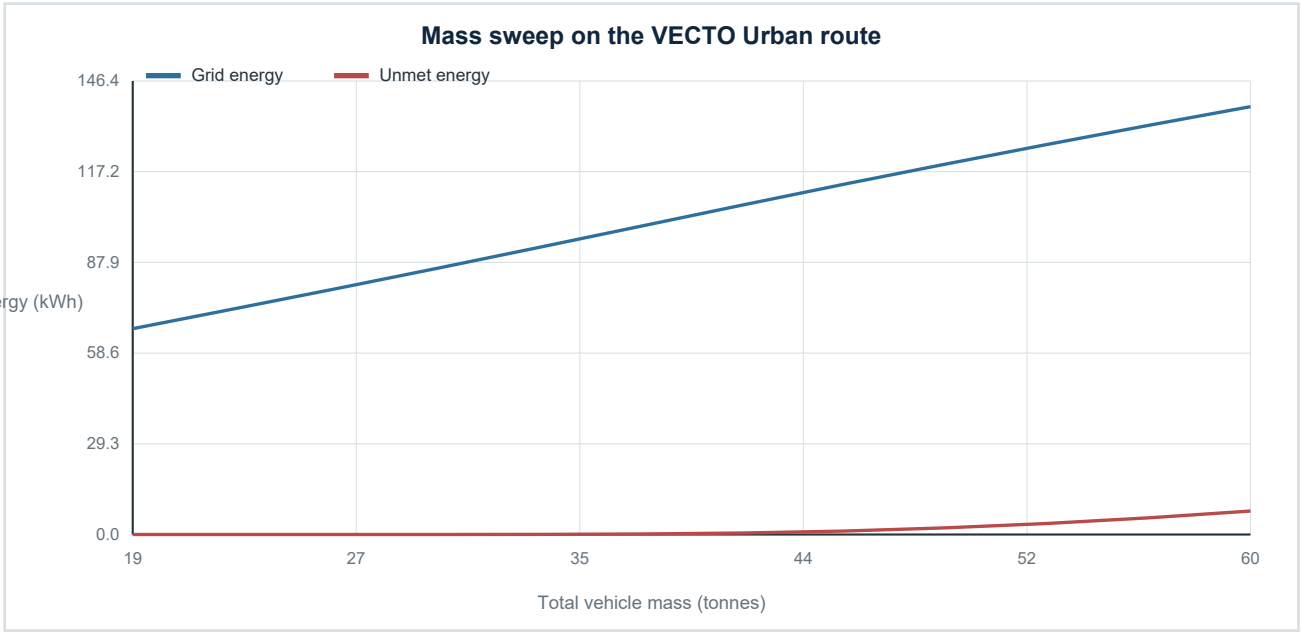


Figure 4.1. Calculated-mass sensitivity from 19 to 60 tonnes by varying entered load

The historical 19-to-60 tonne catalog is retained only as a reproducible sweep grid. Each study point is now realized by calculating the load required above the selected hardware-dependent curb mass. The default component set remains feasible at the low end, but unmet traction energy grows with total mass. Cost per kilometre alone must not rank infeasible cases because unsupplied energy is not purchased.

4.4 Achieved-speed and performance formulations

Backward mode treats route speed as achieved and computes required force. Constrained mode instead limits positive acceleration to the force that remains after road load, motor torque-speed/power, driveline, battery-current, terminal-voltage, and available-energy constraints. Terrain and auxiliary demand are sampled at actual travelled distance, speed is prevented from becoming negative, and a depleted or force-deficient vehicle remains stopped. Performance mode uses a proportional driver, explicit component limits, and a forward longitudinal plant to calculate achieved speed and distance.

$$F_{acc,available} = \max[0, \min(F_{motor}, F_{battery}) - F_{road}] \quad (4.11)$$

$$a_{achieved} = \min[a_{requested}, F_{acc,available}/m] \quad (4.12)$$

$$v_{k+1} = \max[0, v_k + a_{achieved,k}\,\Delta t] \quad (4.13)$$

$$s_{k+1} = s_k + 0.5(v_k + v_{k+1})\Delta t \quad (4.14)$$

Performance evidence
The KPI Vehicle Performance sub-tab reports desired versus achieved speed, route completion, tracking compliance, speed adequacy, RMS/maximum speed error, time below target, limiting cause, and termination reason. Repeat-route studies remain range/depletion experiments rather than single-pass tracking tests.

5 Rear Hub Motors and Fixed Driveline

5.1 Kinematics

$$\omega_{wheel} = v / r_{loaded} \tag{5.1}$$

$$\omega_{motor} = i_{fd}\, \omega_{wheel},\ n_{motor} = 60\, \omega_{motor} / (2\pi) \tag{5.2}$$

The model uses two identical rear hub motors. The selected fixed reduction ratio connects wheel speed to motor speed. If motor speed exceeds the catalog maximum, available motor power is set to zero and the shortfall is logged.

5.2 Torque and power envelope

$$T_{available} = \min[T_{peak},\ P_{peak} / \max(\omega_{motor},\ \varepsilon)] \tag{5.3}$$

$$P_{tractive,max} = 2 \min(P_{peak},\ T_{available}\, \omega_{motor})\, \eta_{fd,mot} \tag{5.4}$$

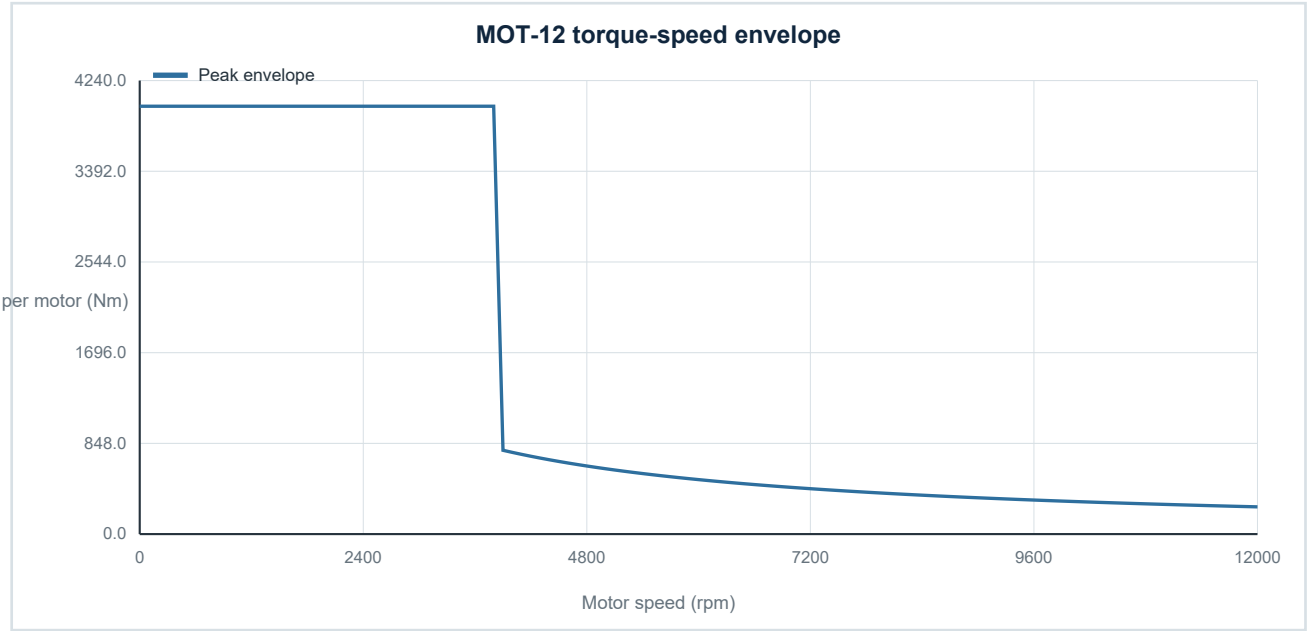


Figure 5.1. Selected motor constant-torque and constant-power regions

5.3 Directional efficiency and regeneration

Traction wheel power is divided by motor and final-drive motoring efficiencies to obtain DC demand. During braking, delivered negative wheel power is multiplied by regeneration efficiencies, producing negative motor DC power. Regeneration is limited by motor capability and fixed-drive direction. The Hybrid limit uses the currently active scaled bank capability; BEV sums the instantaneous capability of every connected parallel pack represented by the set multiplier.

$$P_{motor,dc} = P_{wheel} / (\eta_{fd,mot}\, \eta_{motor,mot}) \text{ for traction} \tag{5.5}$$

$$P_{motor,dc} = P_{wheel}\, \eta_{fd,reg}\, \eta_{motor,reg} \text{ for regeneration} \tag{5.6}$$

6 Battery Energy Model

6.1 Energy state and SOE

Each battery retains usable stored energy as its dynamic state and uses a first-order Thevenin capability layer. Open-circuit voltage, internal resistance, maximum discharge current, and maximum charge current are two-dimensional functions of SOE and ambient battery temperature. This supports state-dependent terminal voltage, current, power capability, and ohmic loss without claiming a full electrothermal cell model.

$$SOE = E / E_{usable} \tag{6.1}$$

$$E_{k+1} = E_k - P_{dis}\,\Delta t / (3600\,\eta_{dis}) \tag{6.2}$$

$$E_{k+1} = E_k + |P_{chg}|\,\eta_{chg}\,\Delta t / 3600 \tag{6.3}$$

6.2 Current, voltage, and energy limits

At every time step the battery helper interpolates the SOE-temperature current maps, OCV map, and resistance map. Map current is further limited by terminal-voltage headroom. Accepted power is then bounded by the resulting electrical capability and by the energy-implied power needed to remain within minimum or maximum SOE.

$$I_{dis,max} = \min[I_{map,dis}(SOE,T),\ (OCV(SOE,T)-V_{min})/R(SOE,T)] \tag{6.4}$$

$$I_{chg,max} = \min[I_{map,chg}(SOE,T),\ (V_{max}-OCV(SOE,T))/R(SOE,T)] \tag{6.5}$$

$$P_{dis,max} = I_{dis,max}[OCV-I_{dis,max}R]/1000 \tag{6.6}$$

$$P_{chg,max} = I_{chg,max}[OCV+I_{chg,max}R]/1000 \tag{6.7}$$

Numerical safeguard
The energy state is clamped to [MinSOE, MaxSOE] after each update. Regenerative power rejected by the active battery is assigned to the resistor load bank; a genset-charger mismatch remains a separately logged rejected-charge term. Unserved positive demand is recorded as unmet DC power.

6.3 Hybrid active/standby roles versus BEV parallel packs

The Hybrid supervisor permits only one battery bank to discharge for traction. This is intentionally conservative: the active bank must support residual DC demand while the equal-sized alternate bank remains available for standby charging and deterministic role switching. Bank energy, mass, charge, discharge, and regeneration limits scale linearly with the whole-number set multiplier. BEV mode removes active/standby roles and connects the installed one-or-more packs in parallel; each selected bank remains independently bounded by SOE, energy headroom, power, derating, and efficiency limits.

Table 6.1. Synthetic battery catalog coverage

Catalog range	Minimum	Maximum
Usable energy	100 kWh	255 kWh
Nominal voltage	600 V	800 V
Mass	850 kg	1450 kg

7 Engine-Generator Set and Fuel Model

7.1 Constant best-efficiency operating point

The genset has one permitted electrical command while on: the catalog OptimumPower_kW value, bounded by the rated maximum. It is not load-following and is not connected to the traction bus. Its dedicated charger sends power only to the pack currently designated standby. Battery charge-power and upper-SOE limits determine accepted power; any transient charger mismatch is recorded separately from regenerative load-bank dissipation.

7.2 Generator and engine maps

$$P_{mech} = P_{gen,dc} / \eta_{gen}(\lambda) \tag{7.1}$$

$$\dot{m}_{fuel} = \max[\dot{m}_{idle}, P_{mech}\, BSFC(\lambda) / (1000\, \rho_{fuel}\, 3600)] \tag{7.2}$$

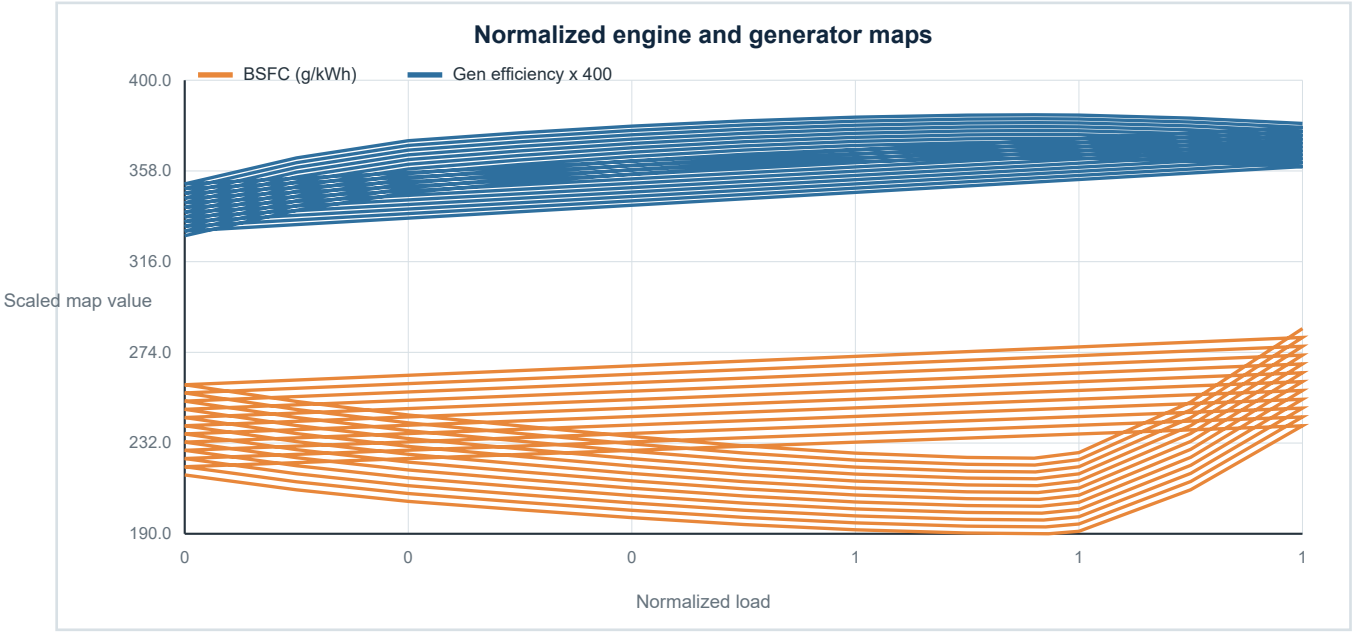


Figure 7.1. Synthetic BSFC and generator-efficiency maps used for interpolation

The generator efficiency is clipped to a physically reasonable numerical range, and BSFC is linearly interpolated over normalized load. A fixed start-fuel penalty is added on every off-to-on transition.

7.3 What the fuel model does not claim

Not an emissions model
There is no engine warm-up, transient combustion, aftertreatment, ambient correction, fuel-temperature effect, or pollutant calculation. The output is a concept diesel-volume estimate, not an emissions or homologation result.

8 Supervisory Energy Management

8.1 Control objectives

In Hybrid mode the deterministic supervisor has five priorities: keep the genset electrically isolated from traction; preserve exactly one active traction pack; swap roles at 30% SOE when the alternate pack is ready; recharge the depleted standby pack at constant best-efficiency genset power; and allocate regeneration strictly to auxiliaries, then the active battery, then the resistor load bank. BEV mode uses the separate policy in Section 8.6.

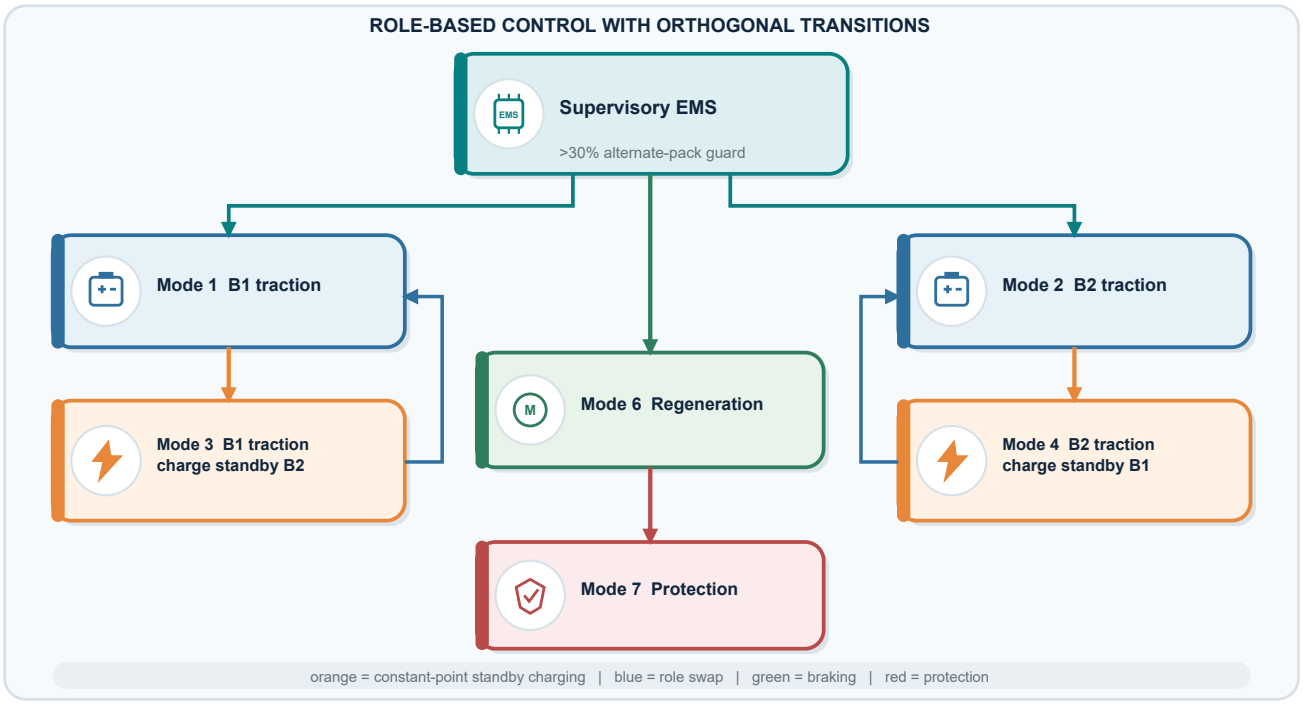


Figure 8.1. Supervisory operating modes and principal transitions

8.2 Decision order

1. Read both pack SOEs and retain the current active-battery state.
2. If active SOE is at or below 30% and the standby pack is above 30%, exchange the active and standby roles immediately.
3. Start the genset when standby SOE is at or below 30%; while on, command the constant OptimumPower_kW value.
4. Send positive motor-plus-auxiliary demand only to the active battery; the genset is excluded from this equation.
5. Send genset power only to the standby battery; the genset never participates in the traction-node balance.
6. For negative motor DC power, supply auxiliary demand first, send the remaining regenerative power only to the active battery, and divert any unaccepted remainder to the resistor load bank.
7. Update energy states, fuel, operating mode, losses, load-bank heat, rejected genset charge, and the balance residual.

8.3 Regeneration allocation by first principles

Let regenerative power be a non-negative available quantity. Auxiliary loads remain powered during braking, so they consume the first portion directly at the traction DC bus. Only the remainder is offered to the active pack. The battery helper then applies its instantaneous charge-power, regeneration-power, efficiency, and upper-SOE limits. The resistor load bank receives exactly the terminal power that the active battery cannot accept. The standby pack is excluded from this path because its only charging source is the isolated genset charger.

$$P_{reg} = \max(0, -P_{motor,dc}) \tag{8.1}$$

$$P_{reg,aux} = \min(P_{reg}, P_{aux}) \tag{8.2}$$

$$P_{reg,rem} = \max(0, P_{reg} - P_{reg,aux}) \tag{8.3}$$

$$P_{dump} = \max(0, P_{reg,rem} - P_{reg,active}) \tag{8.4}$$

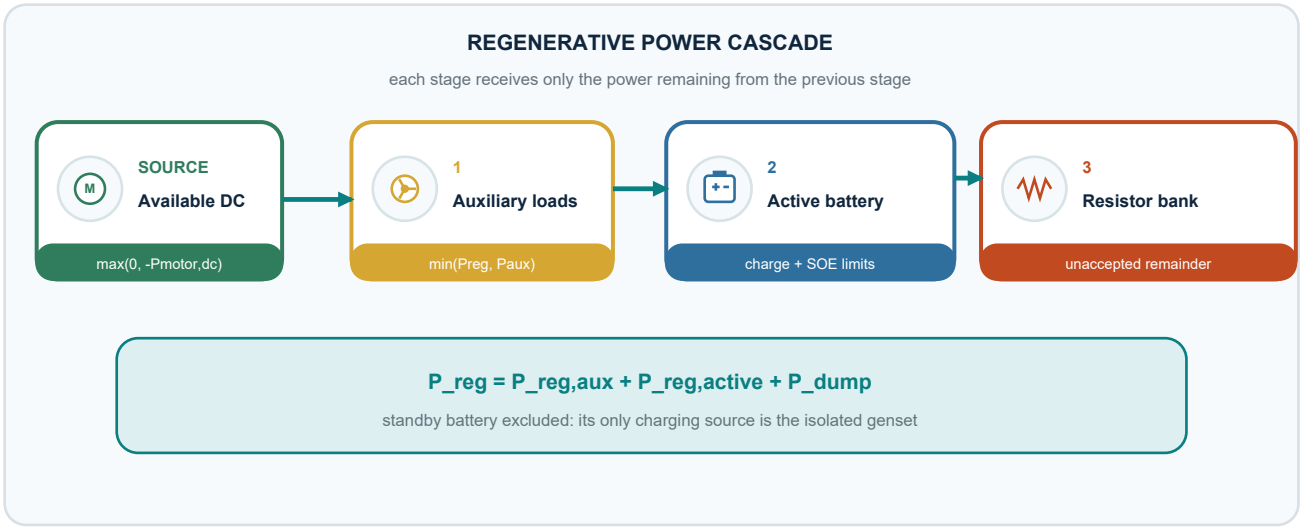


Figure 8.2. Strict auxiliary-active-battery-resistor regeneration hierarchy

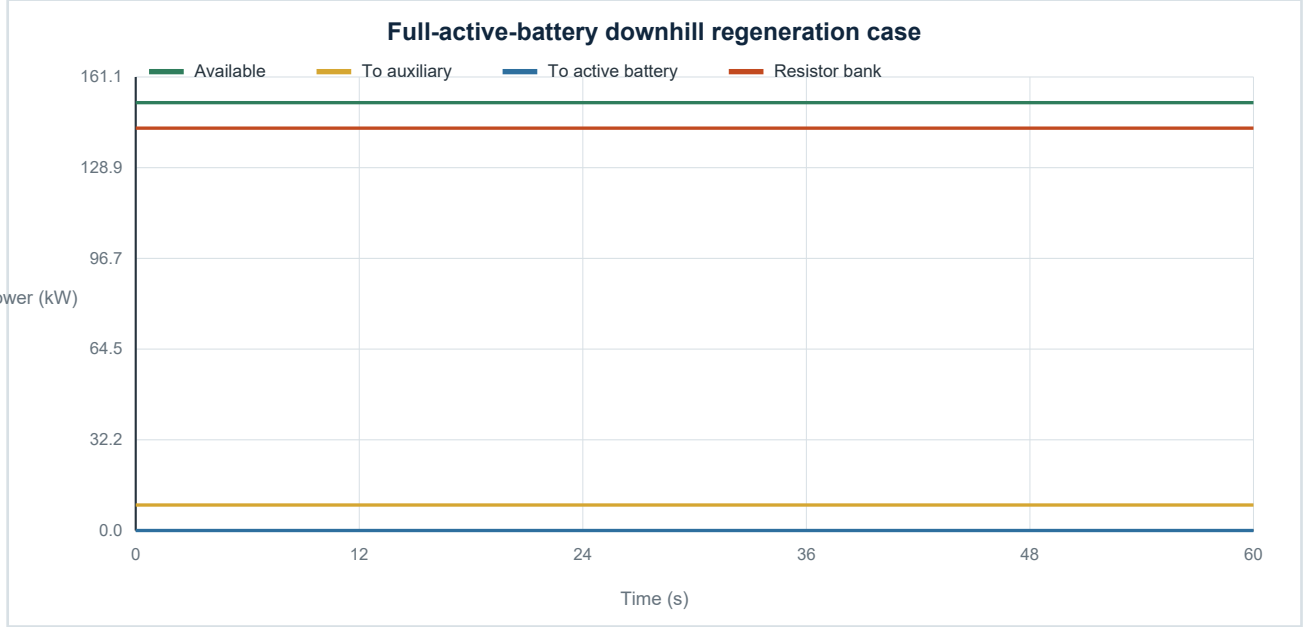


Figure 8.3. When the active battery is full, surplus regenerative power becomes controlled waste heat

Thermal interpretation
The resistor-bank signal is an electrical power sink. A production design would require continuous and transient thermal ratings, cooling airflow, temperature sensing, contactor diagnostics, insulation monitoring, and a braking fallback if the bank is unavailable. Those thermal and safety dynamics are outside the present concept model.

8.4 Hysteresis and dwell time

The 30% role threshold is an architectural rule rather than a tunable calibration. A role swap is allowed only if the alternate pack is above 30%; this prevents rapid swapping when both packs are depleted. If both are low, the current active pack remains connected while the genset charges standby. Charging continues until the standby pack reaches its upper SOE limit or becomes the active pack.

Table 8.1. Operating-mode interpretation

Mode	Meaning	Primary power behavior
1	Battery 1 active	B1 alone supplies positive traction-bus demand
2	Battery 2 active	B2 alone supplies positive traction-bus demand
3	B1 active; B2 charged	Genset fixed output is routed only to standby B2
4	B2 active; B1 charged	Genset fixed output is routed only to standby B1
6	Regeneration	Auxiliary first, active battery second, resistor load bank third
7	Protection	Sources are clamped and unmet demand is recorded
8	BEV single-pack traction	A 0.5 set connects one Battery 1 pack; Battery 2 count is zero
9	BEV parallel traction	All packs represented by the set multiplier share accepted DC power
10	BEV regeneration	Auxiliary first, connected pack(s) second, resistor bank third

8.5 Repeated-route depletion study

The Configuration panel includes a Repeat route until depleted checkbox. When cleared, the solver executes exactly one selected mission, preserving the standard comparison workflow. When selected, the time-speed-grade-auxiliary sequence is concatenated continuously while distance and time remain monotonic. This converts a route case into a range-to-depletion experiment without changing the component physics or supervisory priorities.

The genset consumes no more than the configured fuel-tank volume. After fuel is exhausted, the vehicle continues from available battery energy. The repeated mission terminates only when the active pack reaches its minimum usable-energy bound, the alternate pack is not above the 30% role-swap threshold, and positive DC demand is unmet. Thus the stop condition means that no admissible traction-energy source remains; it is not an arbitrary route-count limit.

$$V_{fuel,used} \geq V_{tank} \tag{8.5}$$

$$E_{active} \leq E_{active,min} \quad \text{and} \quad SOE_{alternate} \leq 30\% \tag{8.6}$$

Interpretation
A repeated-route result is a controlled range estimate for the prescribed repeating mission. It is not a real timetable, charging plan, driver-hours model, thermal endurance result, or warranty prediction.

8.6 BEV supervisory mode

BEV control bypasses the 30% Hybrid role-swap and every genset state. The controller reads deterministic pack counts from the half-step battery-set multiplier, forms total traction-node demand, computes instantaneous scaled-bank capabilities, and distributes accepted discharge or charge power. If requested positive DC power exceeds connected capability, the difference is logged as unmet DC power. During braking, the same auxiliary-battery-resistor priority is applied across every connected pack.

$$P_{BEV,accepted} = \min(|P_{request}|, \Sigma_{j\ connected} C_j) \tag{8.7}$$

$$N_{B1} = \text{ceil}(N_{packs}/2), \quad N_{B2} = \text{floor}(N_{packs}/2) \tag{8.8}$$

BEV invariants
All connected packs start at the same SOE (85% by default). With 0.5 set, Battery 2 count and power are exactly zero. For every BEV multiplier, genset power, fuel rate, fuel consumption, runtime, and starts are exactly zero.

8.7 Ordered BEV-then-Hybrid comparison

A default-off Configuration checkbox provides a controlled two-run experiment. When cleared, Run Manual executes only the architecture selected by the Hybrid-BEV slider. When checked, the orchestrator executes BEV first and Hybrid second with the same route, components, load, prices, auxiliaries, and repeat-route setting. The BEV case makes connected starting SOEs equal to the displayed Battery 1 value; the Hybrid case retains its configured B1/B2 values. Both results remain in memory, Hybrid histories are displayed last, and the status and Simulation Analysis header report both operating-cost outcomes.

Comparable battery-set constraint
The paired run requires a positive whole-number battery-set multiplier because that domain is valid for both architectures. A fractional BEV set is never silently rounded for the Hybrid case; the app reports a clear configuration error instead.

9 Power, Energy, Cost, and Range Accounting

9.1 DC power conservation

Two node equations make the isolation explicit. At the traction node, only the active battery balances motor power, auxiliary demand, and resistor-bank dissipation. At the charger node, the genset balances standby-battery charging and any transient rejected charger power. Summing the nodes gives the project-wide residual without implying a physical genset-to-traction connection.

$$P_{active} - P_{motor,dc} - P_{aux} + P_{unmet} - P_{dump} = 0 \quad (9.1a)$$

$$P_{gen} + P_{standby} - P_{gen,rejected} = 0 \quad (9.1b)$$

$$P_{gen} + P_{B1} + P_{B2} - P_{motor,dc} - P_{aux} + P_{unmet} - P_{dump} - P_{gen,rejected} = P_{residual} \quad (9.1c)$$

$$P_{rejected} = P_{dump} + P_{gen,rejected} \quad (9.1d)$$

$$\Sigma_{j\ connected} P_{Bj} - P_{motor,dc} - P_{aux} + P_{unmet} - P_{dump} = P_{residual} \quad (BEV) \quad (9.1e)$$

9.2 Equivalent grid replenishment

$$E_{grid} = \Sigma_{connected\ packs} \max(0, E_{i,0} - E_{i,f}) / \eta_{grid} \quad (9.2)$$

$$C_{total} = V_{fuel}\, c_{fuel} + E_{grid}\, c_{electricity} \quad (9.3)$$

$$C_{km} = C_{total} / d_{route} \quad (9.4)$$

Terminal battery surplus is not credited by default. This prevents an artificial negative electric cost and makes the equivalent-replenishment comparison conservative.

9.3 Range estimates

$$R_{battery} = E_{usable,initial} / e_{DC,route} \quad (9.5)$$

$$R_{fuel} = V_{tank} / v_{fuel,per\text{-}km} \quad (9.6)$$

Range is conditional
Range estimates assume that the observed route energy intensity and supervisory behavior remain representative. If the example route never starts the genset, the code estimates supported fuel range at the selected optimum genset point rather than reporting infinity.

9.4 Feasibility before economics

A low cost per kilometre is meaningful only when unmet traction energy is negligible and the balance residual satisfies tolerance. An infeasible case can look artificially cheap because energy that the vehicle failed to deliver was never purchased. The optimizer therefore filters feasibility before ranking cost.

10 Backward, Constrained, and Performance Formulations

The three selections answer different engineering questions and therefore use different causal assumptions. They share the same route library, component records, battery and motor maps, powertrain mode, supervisory rules, sign conventions, and result contract. They must not be interpreted as three interchangeable accuracy levels.

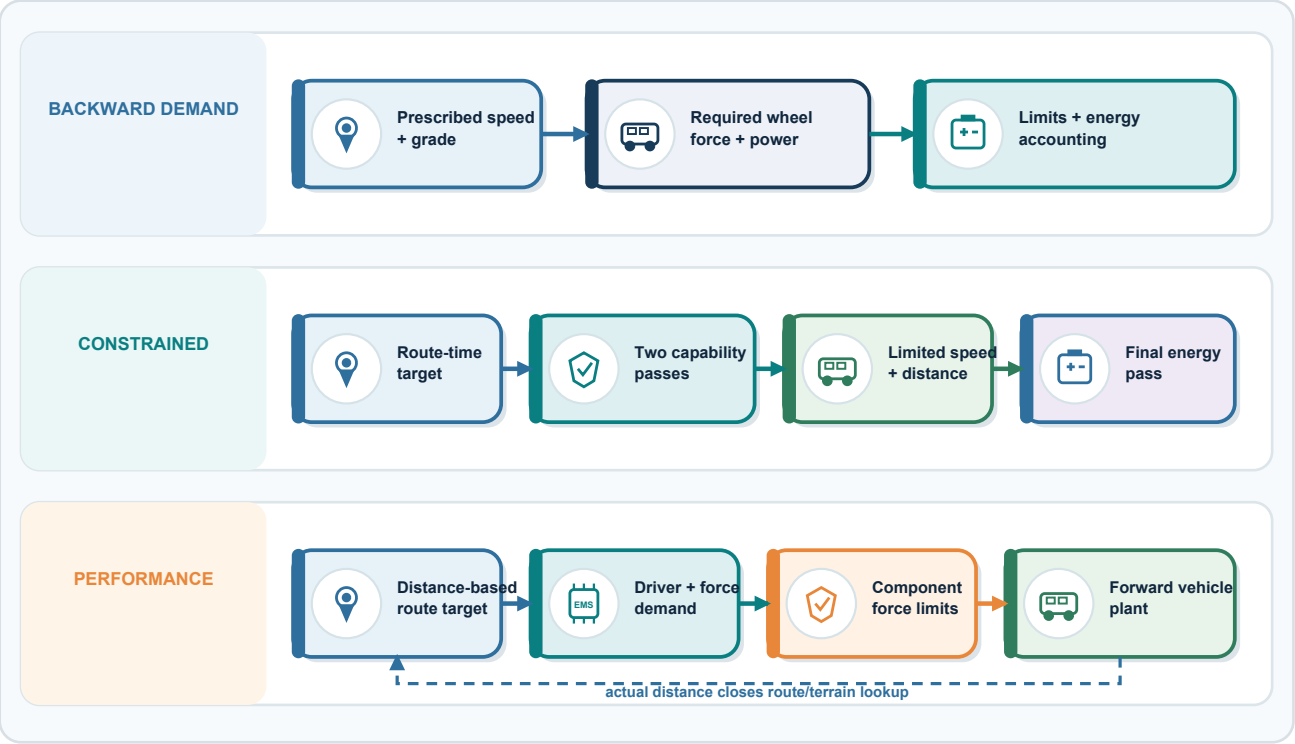


Figure 10.1. Causality and information flow in the three selectable simulation formulations

10.1 Comparison and model-selection guide

Table 10.1. What each formulation calculates and what it can prove

Formulation	Primary question	Motion treatment	Horizon / completion	Implementation artifact
Backward	What energy and component demand is required if the route speed is achieved?	Speed is prescribed; no vehicle-speed feedback state	Exactly the route time trace	Hybrid/BEV SLX models + backward MATLAB core
Constrained	Can available current, torque, power, and force approximately deliver the route within its scheduled time?	Acceleration and achieved speed are suppressed by estimated capability	One fixed route-time horizon; 99.5% distance is the concept completion threshold	Constrained MATLAB function; no separate SLX
Performance	What speed and distance does the vehicle actually achieve under closed-loop demand and physical limits?	Driver, component limits, and longitudinal plant form a forward loop	Continues to route completion, sustained stall, or maximum duration	Performance SLX model + forward MATLAB function

Decision rule
Use Backward for fast energy and rating demand. Use Constrained as a rapid capability screen when the scheduled route time must stay fixed. Use Performance for achieved-speed, gradeability, stall, and mission-completion conclusions. If Backward passes but either achieved-speed formulation fails, the vehicle is energy-accountable but not dynamically capable for that mission.

10.2 Backward demand model

Backward causality begins at the route. At every route-time sample, desired speed is treated as actual vehicle speed. A filtered finite difference gives acceleration; mass, rolling resistance, grade, and aerodynamic drag give required tractive force; and force times speed gives wheel-power demand. The calculation then moves backward through fixed reduction, motor loss and torque-speed limits, the traction DC bus, auxiliaries, battery limits, and the Hybrid or BEV energy manager.

$$v_{actual,k} = v_{route,k} \tag{10.1}$$

$$F_{required,k} = m\, a_{route,k} + F_{roll,k} + F_{grade,k} + F_{aero,k} \tag{10.2}$$

$$P_{wheel,required,k} = F_{required,k}\, v_{route,k} \tag{10.3}$$

Component saturation does not feed back to change route speed. Instead, any motor or battery shortfall is recorded as unmet wheel or DC energy. This is intentional: the model exposes the demand that the selected hardware cannot serve without allowing a lower simulated speed to hide the original requirement. It is therefore excellent for duty-cycle energy accounting and preliminary sizing, but it cannot prove acceleration, hill-climb speed, timetable adherence, or route completion.

Table 10.2. Backward model signal progression

Stage	Principal operation	Evidence produced
Route and environment	Prescribed speed, grade, auxiliary multiplier, ambient and wind	Known mission boundary
Vehicle dynamics	Differentiate speed; sum inertia and road-load forces	Required force, acceleration, wheel power
Hub drive	Apply motor torque-speed/power, loss map, final-drive efficiency, regeneration limit	DC demand, motor loss, unmet motor power, friction braking
Energy management	Route traction/auxiliary demand; apply battery limits; switch Hybrid roles; charge standby only	Battery power/SOE, genset power, regen allocation, load-bank heat
Accounting	Integrate electrical energy, fuel, cost, and balance residual	Energy/cost KPIs and feasibility warnings

Simulink structure
HybridBus_BackwardModel.slx contains Route and Environment, Vehicle Longitudinal Dynamics, Rear Hub Motor Drive, Auxiliary Loads, Supervisory Energy Management, two Battery Packs, Engine Genset, Energy and Cost Accounting, and Output Logging. HybridBus_BEVModel.slx preserves the demand-side chain but replaces the Hybrid supervisor and genset with parallel BEV battery energy management.

10.3 Fast constrained formulation

Constrained mode was added to answer a practical question faster than the detailed forward simulation: if the scheduled route remains one fixed time trace, how much of it can the selected vehicle physically cover? The algorithm uses the route speed as a driver target, but it integrates a separate achieved-speed state. At each sample it calculates road load at actual travelled distance, determines the requested acceleration from speed error, and caps positive tractive force by the lesser of motor and battery capability after auxiliary demand.

$$a_{requested} = clip[K_p(v_{target}-v), -a_{dec,max}, a_{acc,max}] \quad (10.4)$$

$$F_{available} = \min(F_{motor\ limit}, F_{battery\ limit}) \quad (10.5)$$

$$F_{delivered} = \min(F_{demand}, \max[0,F_{available}]) \quad (10.6)$$

$$a_{achieved} = (F_{delivered}-F_{road\ load})/m \quad (10.7)$$

The implementation deliberately performs two capability/speed iterations and then one final energy-consistent backward-kernel pass. The first pass estimates capability from an initial achieved-speed trace; the second updates battery state and speed-dependent limits; the final pass recomputes energy and power histories on the converged approximate trace. Grade and auxiliary demand are sampled at actual distance, speed and distance cannot decrease, and a depleted or force-deficient vehicle remains stopped.

1. Start with zero achieved speed over the original route-time vector.
2. Build a temporary route from the current achieved speed and run the backward energy kernel.
3. Use battery discharge capability, auxiliary demand, motor torque-speed limits, final-drive force, and road load to integrate a limited speed and distance trace.
4. Repeat the capability update once, then execute a final backward energy pass on the updated trace.
5. Report route completion, distance shortfall, maximum achieved speed, RMS and maximum speed error, time below target, dominant limiting cause, and termination reason.

Why there is no Constrained .slx file
Constrained is a MATLAB algorithmic wrapper around the validated backward energy kernel, not a third standalone Simulink model. It is an iterative quasi-forward approximation designed for bounded runtime. Repeat route until depleted is disabled in this mode, and failure to finish before the original route-time horizon is reported as incomplete rather than extending the schedule.

10.4 Forward Performance model

Performance mode reverses the causality used by Backward mode. Route speed becomes a target rather than an imposed state. Actual distance addresses the route, terrain, auxiliary, and station-stop data; a proportional driver converts speed error into bounded requested acceleration; the force-demand block adds road load; component limits determine delivered traction or regenerative braking; and the vehicle plant integrates the resulting net force into actual acceleration, speed, and distance.

$$e_{v,k} = v_{target}(s_{actual,k}) - v_{actual,k} \quad (10.8)$$

$$F_{demand,k} = m\, a_{requested,k} + F_{road\ load,k} \quad (10.9)$$

$$F_{drive,k} = \min(F_{demand,k}, F_{motor,k}, F_{battery,k}) \quad (10.10)$$

$$v_{k+1} = \max[0, v_k + (F_{drive,k}-F_{road\ load,k})\Delta t/m] \quad (10.11)$$

During braking, the motor regeneration envelope supplies as much negative wheel force as speed and component limits allow; pneumatic/friction braking supplies the remainder. On the DC side, regeneration still follows the same priority: auxiliaries first, active or connected battery capacity second, and resistor load bank third. Hybrid role switching and standby-only constant-point genset charging, or BEV parallel battery sharing, are evaluated at every forward time step.

Table 10.3. Forward Performance signal loop

Subsystem / function	Inputs	Outputs and role
Route Target and Terrain	Actual distance	Desired speed, grade, auxiliary multiplier; handles stop approach and dwell
Driver and Force Demand	Desired speed, actual speed, grade	Speed error, bounded desired acceleration, total force demand
Component Force Limits	Force demand, speed, battery available power	Delivered force and unmet force after motor/battery constraints
Forward Vehicle Plant	Delivered force and grade	Nonnegative actual speed, actual distance, actual acceleration
MATLAB energy loop	Plant motion plus component states	Battery/genset/fuel, regen/friction braking, limits, costs, and conservation

Unlike Constrained mode, Performance may continue beyond nominal route time while attempting to reach the route endpoint. It terminates when the route is completed, when sustained inability to overcome road load satisfies the stall rule, or when the maximum allowed simulation duration is reached. This makes it the appropriate formulation for gradeability, acceleration adequacy, route completion, and stopped-vehicle conclusions, while still remaining a concept model until driver, tyre, thermal, and component data are calibrated against a real bus.

10.5 Shared discretization and result contract

All formulations use the configured fixed sample time, normally 1 s. Energy updates integrate signed power in kWh, fuel rate integrates in litres, and every result uses the same Metadata, SelectedConfiguration, InputParameters, Signals, Summary, and Validation groups. This common contract allows the app to reuse Signals, Detailed Plot, Simulation Analysis, and KPI views without pretending that prescribed and achieved motion have the same evidential meaning.

Table 10.4. How to interpret common outputs without mixing model causality

Reported quantity	Backward interpretation	Constrained / Performance interpretation
Vehicle speed	Prescribed route speed	Achieved state after physical force and power limits
Vehicle distance	Prescribed route integration	Actual nondecreasing travelled distance
Unmet energy	Demand the selected hardware could not supply at prescribed speed	Residual shortfall after achieved motion has already been limited
Route completion	Not a dynamic proof	Explicit distance-based completion result
Speed error	Zero by construction; not performance evidence	Driver target minus achieved speed
Limiting cause	Component saturation in demand accounting	Dominant cause of actual speed or distance shortfall

10.6 Execution sequence

```
cd('C:\TempData\Hybrid_Vehicle\HybridBusProject')
project = startup_hybrid_bus;
db = load_hybrid_bus_database(fullfile('data','HybridBus_ComponentDatabase.xlsx'));

backward = run_hybrid_bus_simulation('', struct('SimulationFormulation',"BackwardDemand"));
constrained = run_hybrid_bus_simulation('', struct('SimulationFormulation',"ConstrainedBackward"));
performance = run_hybrid_bus_simulation('', struct('SimulationFormulation',"ForwardPerformance"));

open_system('HybridBus_BackwardModel.slx')
open_system('HybridBus_PerformanceModel.slx')
app = HybridBusApp;
```

The dispatcher selects simulate_hybrid_bus_core, simulate_hybrid_bus_constrained, or simulate_hybrid_bus_performance from Input.SimulationFormulation. The app exposes the same choice as Backward, Constrained, and Performance. The runner loads and validates the database, resolves a complete input structure, executes the selected kernel, attaches validation metadata, and optionally exports MAT and CSV files.

10.7 Results structure

Table 10.5. Simulation output contract

Field	Contents
Metadata	database file/version, MATLAB release, model version, timestamp
SelectedConfiguration	stable IDs for route and every selected component/calibration
InputParameters	fully resolved scalar structures, maps, and resampled route
Signals	grouped vehicle, wheel, motor, auxiliary, regeneration, battery, genset, controller, and energy histories
Summary	distance, fuel, grid energy, regeneration allocation, load-bank heat, costs, range, SOE, unmet energy, residual, and mass decomposition
Validation	warnings, constraint violations, and feasibility flag

10.8 Verification evidence and recommended comparison

Table 10.6. Formulation-specific automated evidence

Formulation	Automated evidence	Principal acceptance statement
Backward	Constant-speed hand calculation; acceleration and grade response; saturation; conservation; repeatability	Required power follows first-principles equations and all unsupplied demand remains visible
Constrained	Acceleration suppression; depletion stop; distance-indexed terrain; Hybrid/BEV; fixed horizon	Limited hardware lowers achieved speed/distance without rollback and preserves energy balance
Performance	Flat-route completion; severe-grade stall; distance-indexed terrain; Hybrid/BEV; dispatcher; SLX structure	Forward motion completes capable cases and terminates incapable cases without crediting prescribed distance

For a defensible vehicle decision, run all three formulations with identical route, powertrain, component IDs, load, auxiliaries, prices, initial SOEs, and battery-set multiplier. First use Backward to establish the requested energy and peak demand. Next use Constrained to identify rapid capability shortfalls within the scheduled horizon. Finally use Performance to confirm achieved speed, completion time, gradeability, stopping behavior, and stall outcome. Differences between the results are expected evidence of their different causality, not automatically a software defect.

Credibility boundary
The automated tests verify implemented equations, interfaces, limits, invariants, and deterministic behavior. They do not replace supplier-map calibration, driver and tyre validation, thermal and ageing models, or measured CAN/GPS/dynamometer comparison. Performance conclusions remain concept-level until those evidence gaps are closed.

11 Using the Explorer App

11.1 Configuration panel

The programmatic UIFigure app groups route, editable load, calculated curb and total mass, and auxiliary selection in Mission Inputs. Battery, motor, and genset selections remain directly below. Initial battery SOEs are entered as percentages and converted to fractions only at the model boundary. Fuel and electricity prices show EUR/L and EUR/kWh. The battery-set multiplier controls installed bank count; Repeat route until depleted selects a range experiment; and Run BEV first, then Hybrid selects the ordered architecture comparison defined in Section 8.7.

The Powertrain Architecture tab opens first and includes a blue/gray Hybrid-BEV switch. In Hybrid it shows the isolated standby-charging chain, active-battery selector, traction DC bus, regenerative return, auxiliary-first branch, and resistor load bank. In BEV it replaces the fuel and genset chain with external charging and scalable parallel battery banks. Every component block is clickable: a non-modal window reports selected catalog IDs, ratings, units, implemented role, control rules, mass contribution, and concept-model limitations.

> **Current app boundary**
> The current tabs are Powertrain Architecture, Route Map, KPIs, Simulation Analysis, Model Credibility, Signals, and Detailed Plot. Configuration provides Run Manual, Cancel, and Export. The former Optimize control, Max configurations field, and Optimization Ranking tab have been removed. Optional catalog screening remains available only through the headless MATLAB API in Chapter 12.

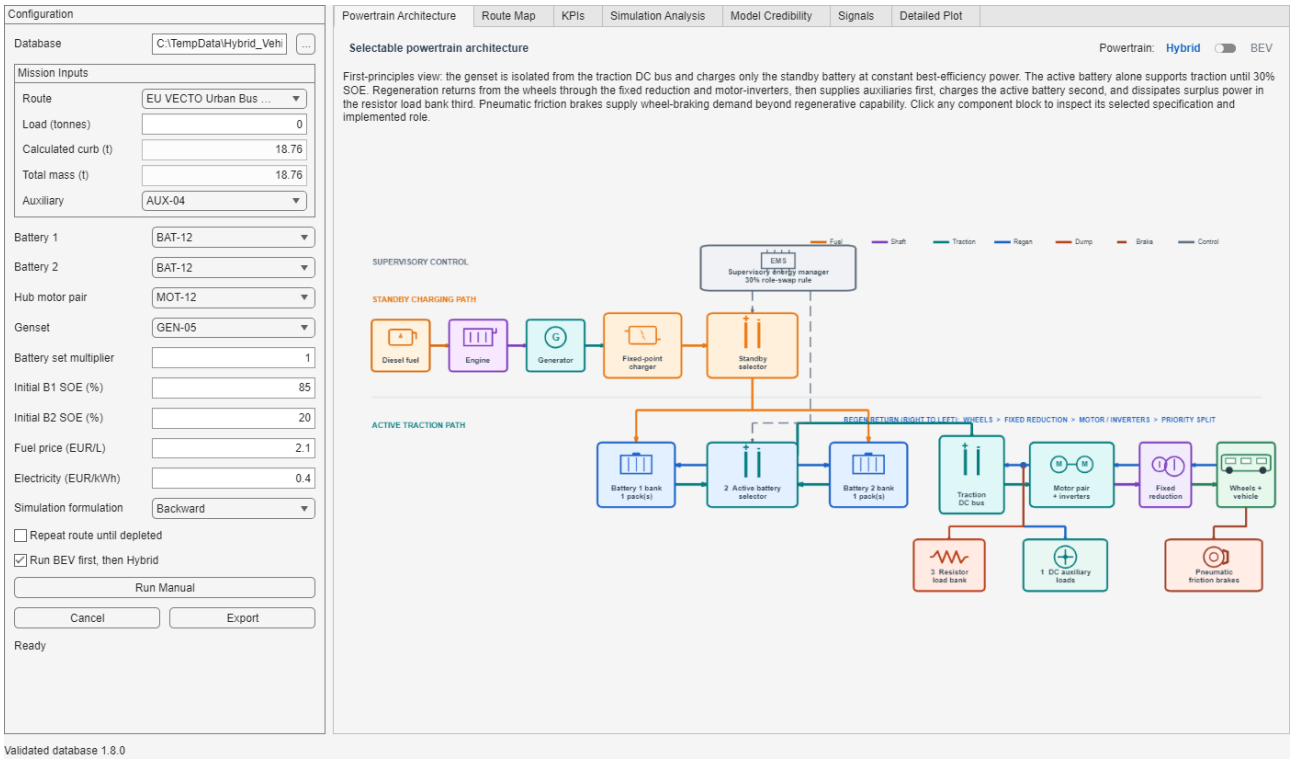


Figure 11.1. Current Explorer Configuration panel with calculated mass and the default-off BEV-then-Hybrid comparison control

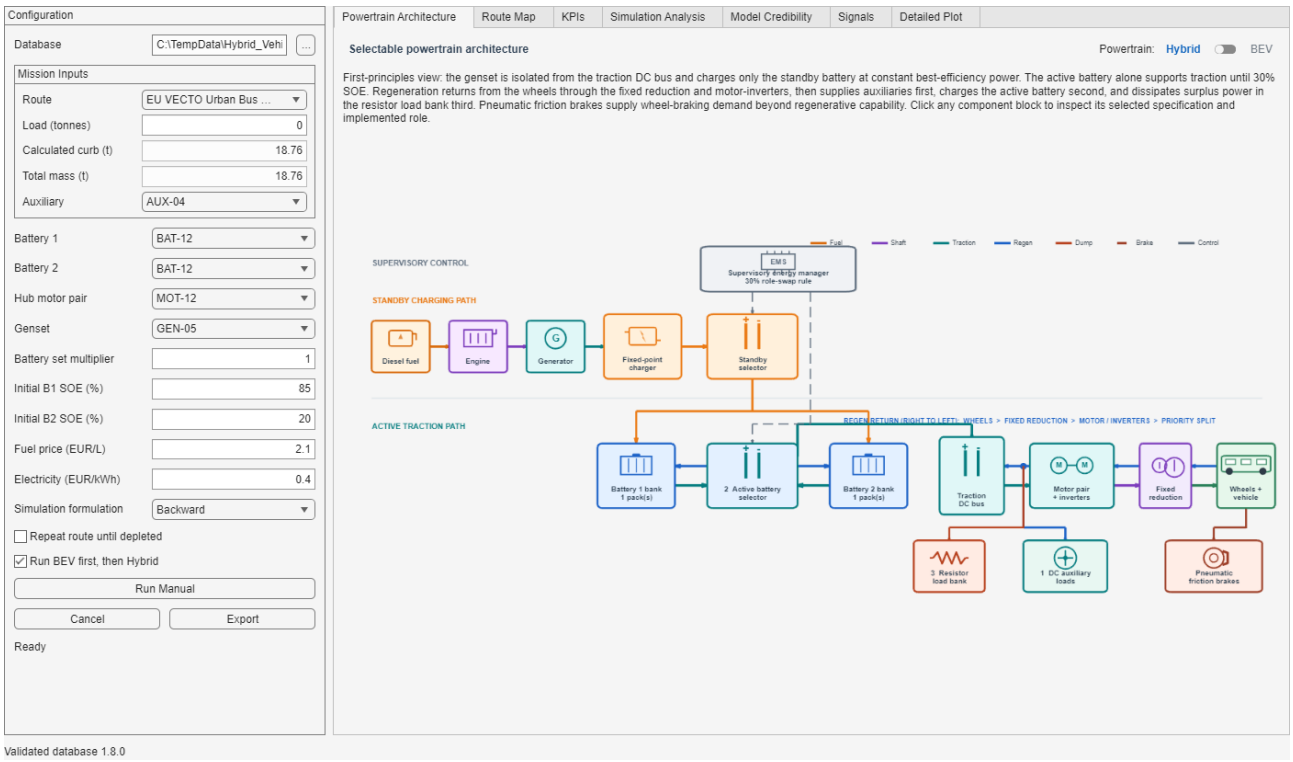


Figure 11.2. Architecture tab with the wheel-to-inverter regenerative return and numbered destination priorities

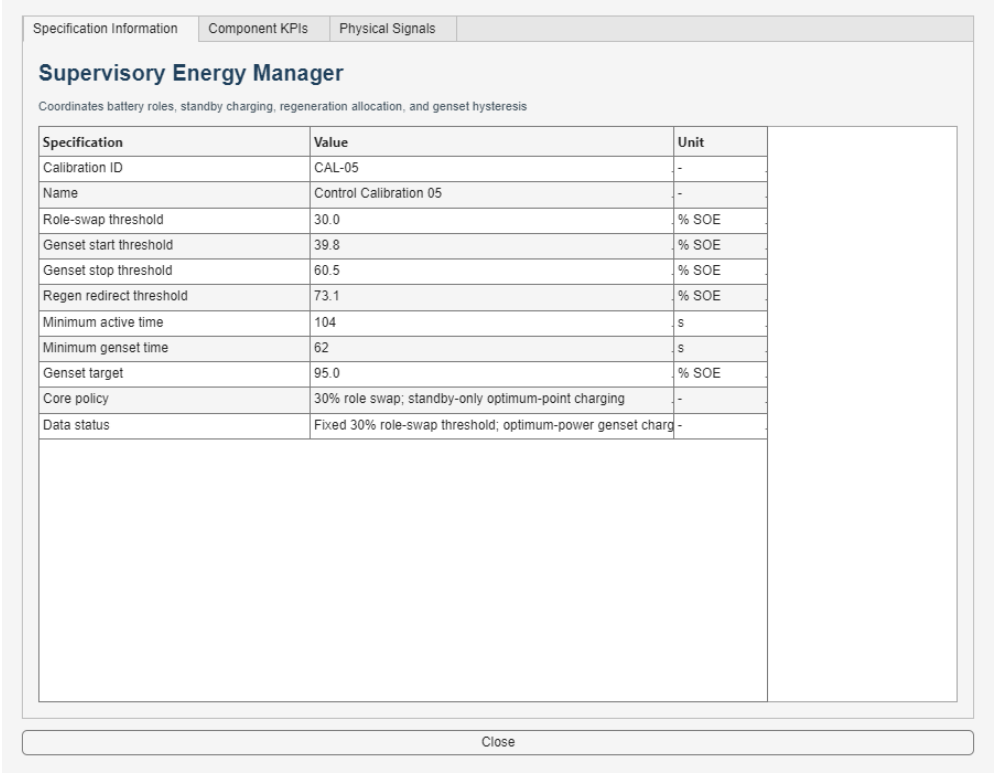


Figure 11.3. Component detail window with Specification Information, KPIs, and Physical Signals tabs

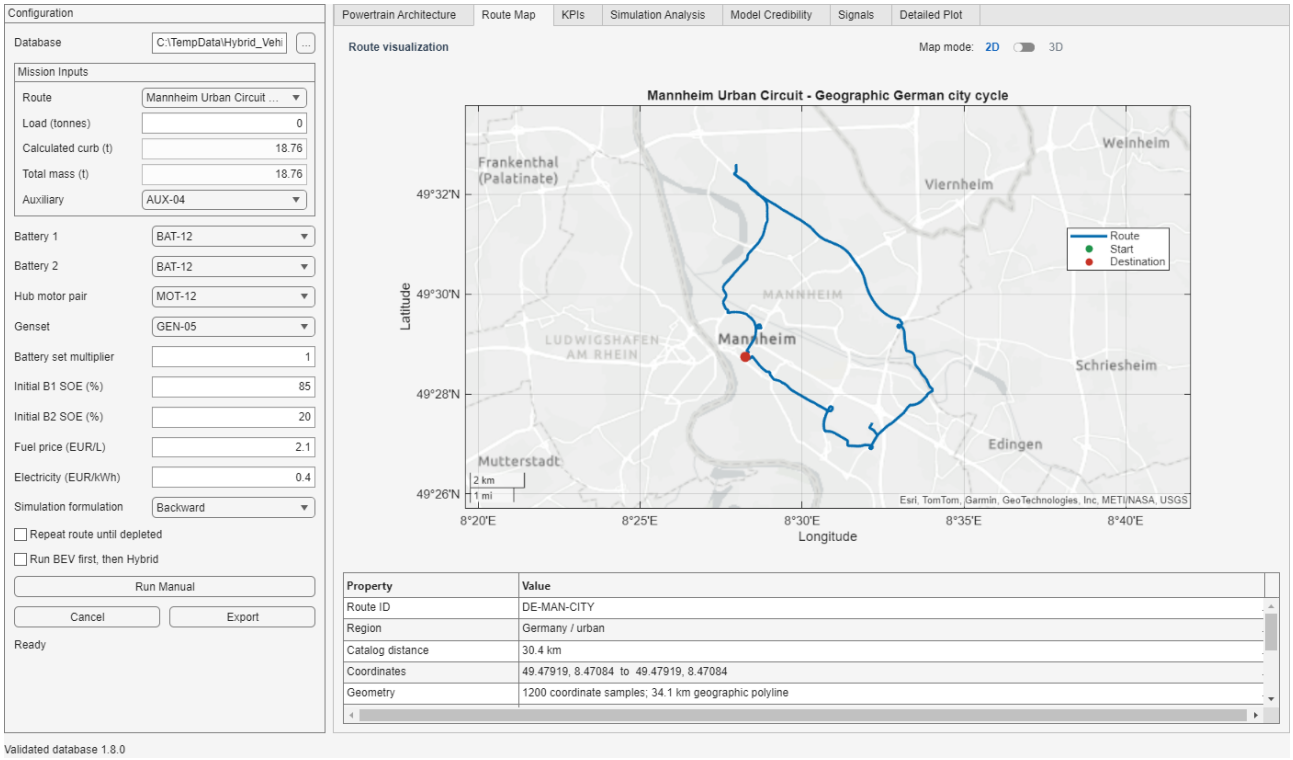


Figure 11.4. Route Map 2D view displaying stored latitude/longitude geometry and endpoint markers

The Route Map switch preserves the geographic 2D view and reveals a second Elevation-Slope switch only in 3D mode. Elevation is plotted in metres from the stored route geometry. Slope is derived from successive elevation change divided by geographic path-distance change and displayed in percent. Blue identifies the selected side of every slider-style switch; gray identifies the alternative.

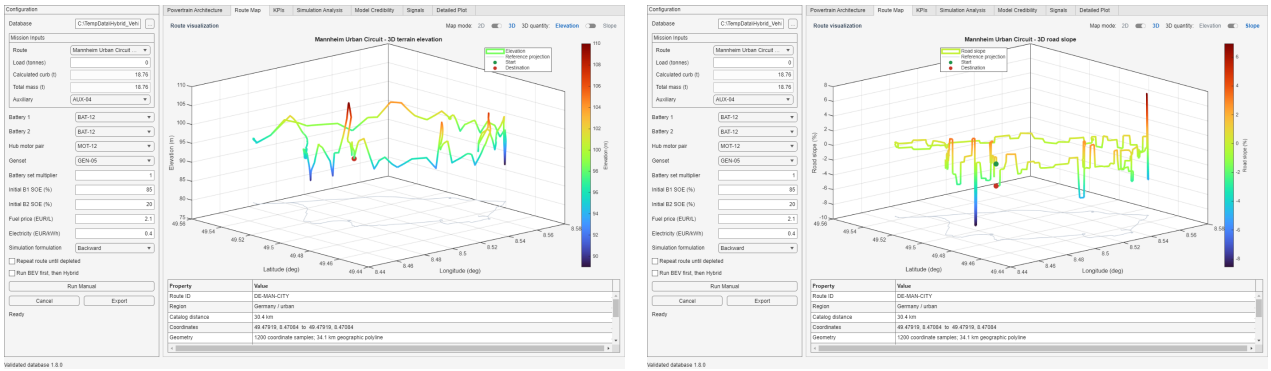


Figure 11.5. Route Map 3D quantity switch: elevation in metres (left) and road slope in percent (right)

11.2 Manual-study and architecture-comparison workflow

1. Select the database and confirm that the status reports a validated version.
2. Choose the route, load in tonnes, auxiliary profile, components, battery-set multiplier, initial SOEs, and prices; confirm the calculated curb and total masses.
3. Leave Repeat route until depleted clear for a one-cycle comparison, or select it deliberately for a range-to-depletion study.
4. Leave Run BEV first, then Hybrid clear to execute the architecture selected by the slider, or check it to execute the ordered paired comparison.
5. Run the case and inspect speed, power flow, battery SOE, genset/fuel, KPIs, Simulation Analysis, and warnings. After a paired run, the plots show Hybrid while the header preserves both cost outcomes.

6 Treat any unmet traction energy as a failed design case, not as an acceptable low-cost solution.

7 Export the selected result only after checking assumptions and feasibility.

Table 11.1. Default ordered BEV-then-Hybrid manual comparison

Mode	Mass (t)	Cost (EUR/km)	Fuel (L)	Grid eq. (kWh)	Feasible
BEV	17.900	0.6492	0.00	64.58	Yes
Hybrid	18.764	4.4411	71.56	66.06	Yes

Reading the paired result
This is a controlled architecture comparison, not an optimization. Component IDs and mission inputs are held fixed, but architecture-specific mass and starting-SOE rules still apply. The BEV excludes genset mass and fuel; Hybrid includes the selected genset and active/standby starting states.

11.3 Management KPI dashboard

The KPIs tab is organized into Executive Decision, Engineering Scorecard, Vehicle Performance, and Robustness and Range sub-tabs. Executive Decision compares BEV and Hybrid outcomes for the selected route, load, auxiliaries, prices, and run scope and provides a qualified recommendation. Engineering Scorecard exposes completion, energy, power, SOE, conservation, and cost gates. Vehicle Performance distinguishes prescribed demand from achieved-speed evidence. Robustness and Range compares operating cost, source energy, range/depletion behavior, and decision sensitivity. Single-mode, ordered BEV-then-Hybrid, and repeat-until-depleted runs are interpreted without inventing unavailable comparison evidence.

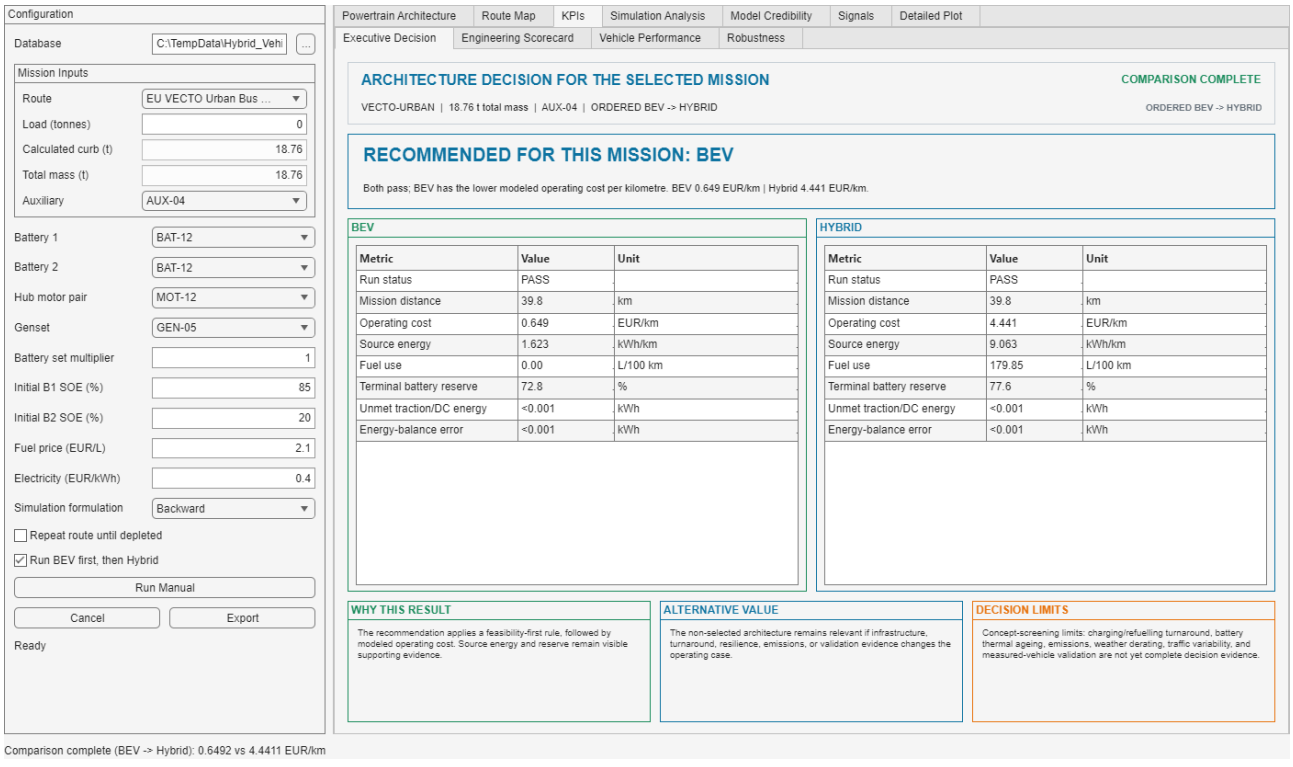


Figure 11.6. Management KPI dashboard with icon-led values, units, configuration context, and feasibility status

11.4 Simulation Analysis tab

Simulation Analysis is an interpretation layer, not another solver. Its energy-allocation chart compares grid-equivalent energy, genset electrical output, recovered regeneration, auxiliaries, battery throughput, and load-bank waste. The duty-share chart reports the fraction of simulated time for which each battery held the active traction role. The assessment table converts principal limits and balances into concise engineering statements covering mission completion, SOE windows, regeneration utilization, dump energy, genset runtime and starts, unmet demand, and conservation error.

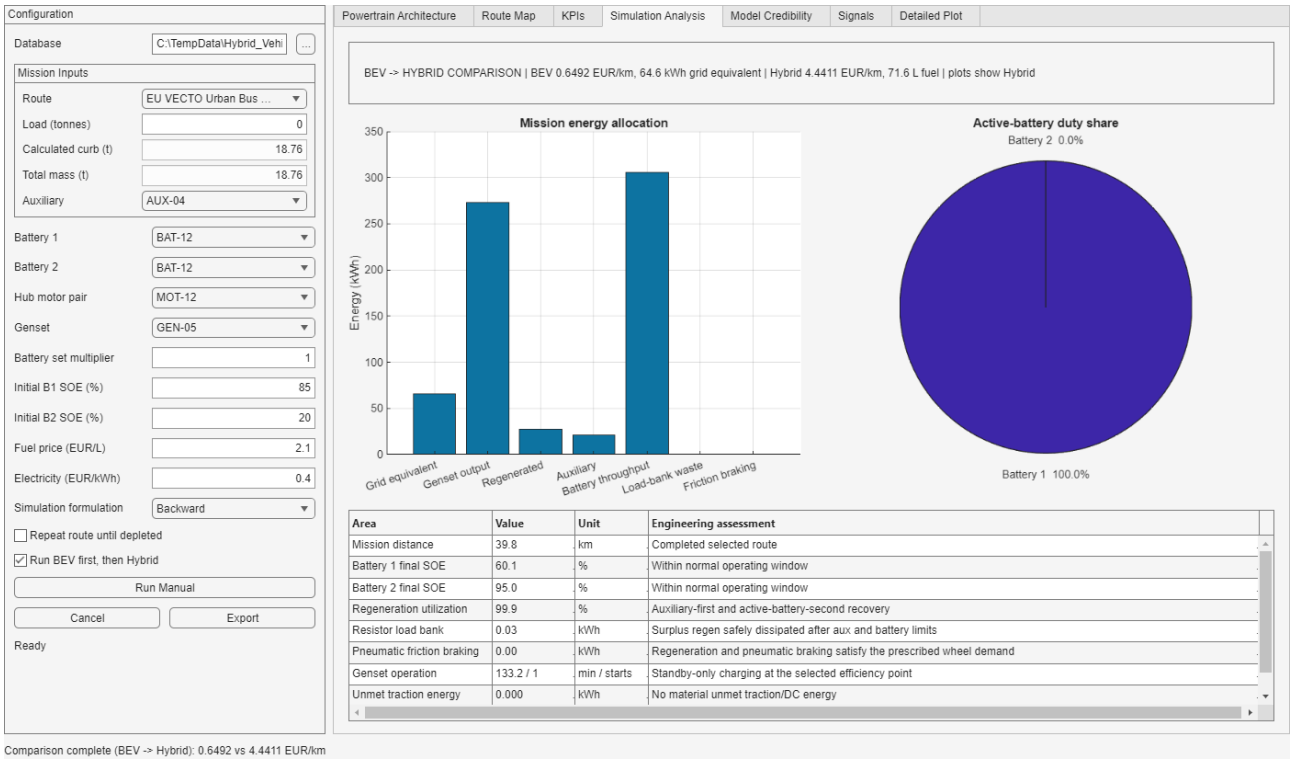


Figure 11.7. Simulation Analysis tab combining energy allocation, battery duty share, and engineering assessment

> **Management versus engineering views**
> Use the KPI dashboard to communicate outcomes quickly. Use Simulation Analysis to explain why those outcomes occurred and whether a limit, energy sink, or control transition requires attention.

11.5 Model Credibility tab

The Model Credibility tab prevents evidence from being compressed into one misleading green status. Concept verification, implementation equivalence, supplier calibration, and measured-vehicle validation are separate gates. Requirements traceability, the behavioral suite, deterministic sensitivity screening, and both Hybrid and BEV MATLAB-Simulink equivalence currently pass, while supplier calibration and measured-vehicle validation remain unavailable. The tab keeps every PASS, FAIL, NOT AVAILABLE, and NOT IMPLEMENTED state visible with its evidence and required decision.

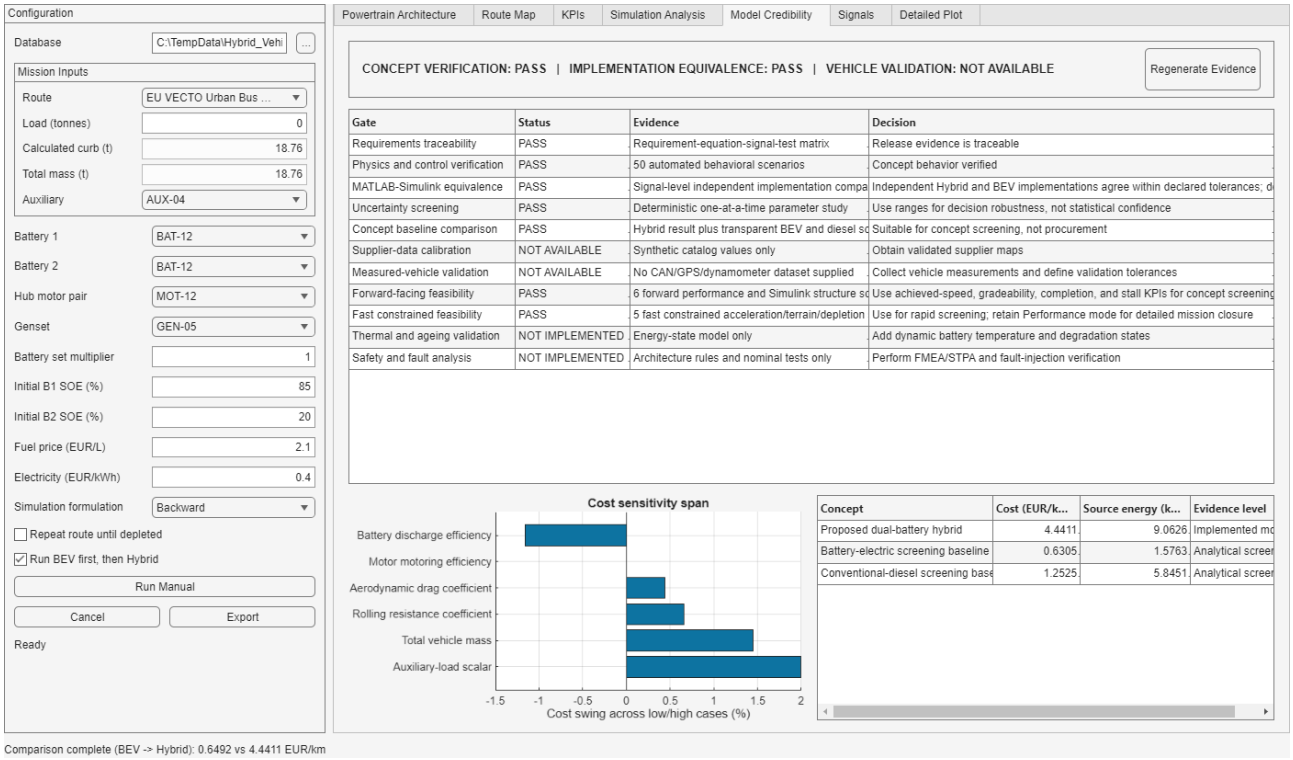


Figure 11.8. Model Credibility tab separating verified concept behavior from open implementation and validation gates

11.6 Signals and Detailed Plot axes

Signals and Detailed Plot share a Time-Distance switch. Time remains the internal independent variable in seconds, but display units adapt to mission duration: minutes for missions shorter than two hours and hours for longer missions. Distance mode uses cumulative vehicle distance in kilometres, so users can relate power, SOE, torque, and speed events directly to route position. The Battery SOE axes are always displayed in percent.

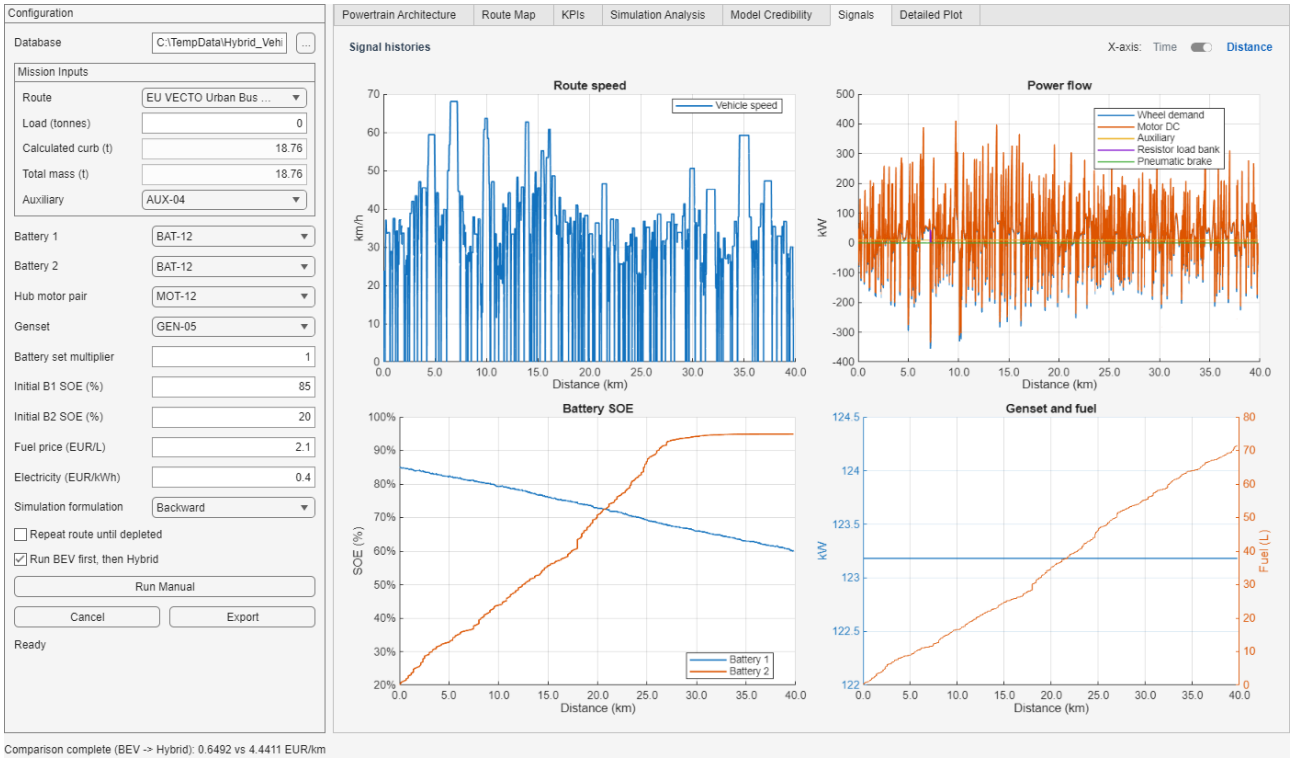


Figure 11.9. Signals tab using cumulative distance in kilometres as the common x-axis

12 Headless Configuration Screening

12.1 Purpose and study boundary

The retained optimize_hybrid_bus_configuration MATLAB function is a deterministic, script-driven design-screening tool. It is not exposed by the current Explorer app. It answers: among the component combinations actually evaluated for one prescribed mission, which feasible combination has the lowest modeled operating cost per kilometre? It does not redesign components, tune continuous parameters, predict hardware life, or prove a global optimum.

Table 12.1. Study variables supplied to the headless screening API

Mode / held fixed by the calling script	Varied by the headless API
Both: route, entered load, calculated mass, auxiliaries, environment, prices	Both: motor pair and final drive
Both: tyre, control calibration, fuel tank, repeat-route flag	Hybrid: Battery 1 and genset
Both: battery-set multiplier; Hybrid: Battery 2 and both initial SOEs	BEV: Battery 1 and Battery 2
BEV: equal initial SOE and connected pack count derived from multiplier	BEV: no genset candidate

Mode-specific search
Hybrid preserves the selected standby Battery 2 while varying Battery 1. BEV varies both pack selections because both may operate in parallel; the voltage-class compatibility rule is enforced before simulation.

12.2 Candidate set and mixed-radix enumeration

For every varied category, the optimizer reads stable component IDs from catalog rows marked OptimizationEnabled. If the four candidate counts are nB, nM, nG, and nF, the complete discrete search space is

$$N_{space} = n_B\, n_M\, n_G\, n_F \tag{12.1}$$

A mixed-radix counter converts each linear evaluation index into one Battery 1, motor, genset, and final-drive combination. This avoids materializing a large Cartesian-product table in memory and gives repeatable ordering between runs with the same database.

Before enumeration, any preferred component ID supplied by the calling script is moved to the front of its candidate list. The first candidate is therefore the complete preferred configuration, including the selected final drive. The complete configuration handoff includes tyre, final drive, environment, control calibration, initial active battery, auxiliary scalar, and fuel-tank capacity.

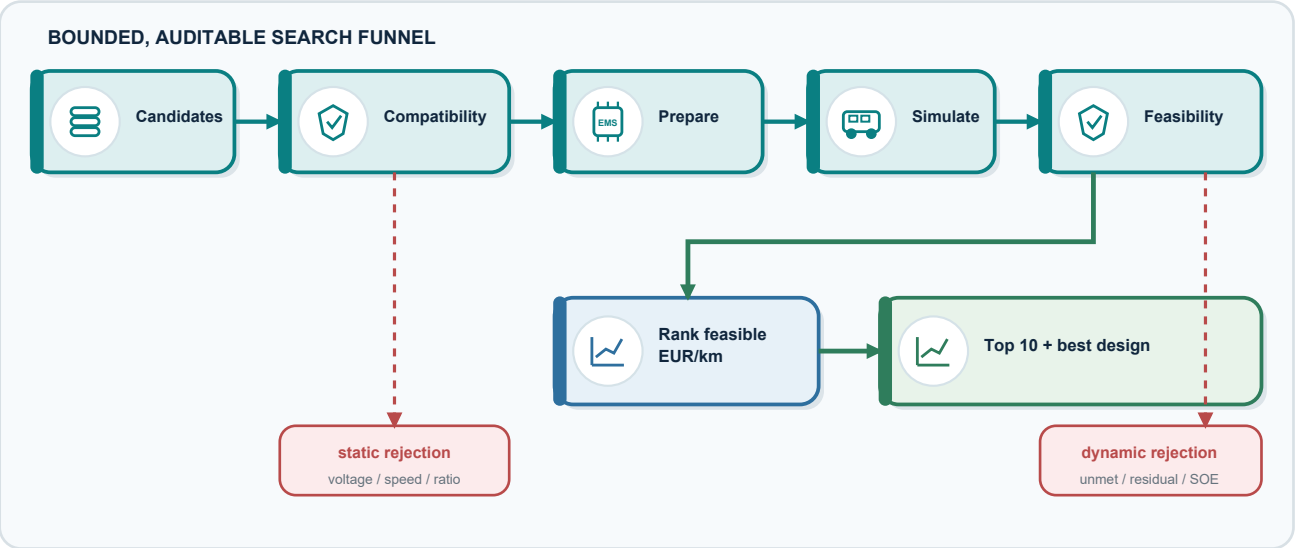


Figure 12.1. Compatibility, simulation, feasibility, and ranking sequence

12.3 Meaning of the evaluation budget

The MaxConfigurations API option is the evaluation budget for one script-driven screening run. A value of 40 means that no more than forty candidate combinations are inspected. Both compatible simulated cases and combinations rejected by the pre-simulation compatibility filter consume an evaluation number. It is a runtime-control argument, not an app field and not a request for forty feasible solutions.

$$N_{evaluated} = \min(N_{max}, N_{space}) \tag{12.2}$$

Increasing the budget explores more of the deterministic catalog order and may find a lower-cost feasible design, but simulation time rises approximately with the number of compatible cases. If Nmax is smaller than Nspace, the reported winner is only the best among the evaluated prefix. A global discrete optimum is established only when the complete enabled search space is evaluated, or when a separately justified search strategy covers it.

Table 12.2. How to interpret the evaluation budget

Setting	Interpretation	Expected effect
Small budget	Rapid screening of the first enabled combinations	Fast, but stronger risk of missing a better design
40 (default)	At most forty candidates are checked	Practical interactive study size
At least Nspace	Every enabled combination is checked	Exhaustive discrete ranking for the stated catalogs

12.4 Compatibility filters

Compatibility filtering removes combinations that violate static interface or rating constraints before the drive-cycle simulation. Rejected rows remain in the evaluated-results table with a reason, making the search auditable.

- Battery-to-motor DC voltage class difference no greater than 50 V.
- Motor speed below maximum at the selected route speed, tyre radius, and final-drive ratio.
- Final-drive ratio inside the selected motor compatibility interval.
- Hybrid: at least one selected battery has adequate continuous discharge rating for the motor rating.
- BEV: summed connected-pack discharge capability covers the pair's continuous motor rating.
- Hybrid only: genset optimum power is within its rating and within either standby battery's charge-power capability.

12.5 Simulation and feasibility gates

Every compatible row is resolved into a complete model input and simulated over the selected route. A low cost is not sufficient. The row enters the feasible ranking only if it passes all three gates below.

$$E_{unmet,traction} < 10^{-3}\ kWh \tag{12.3}$$

$$E_{balance,error} \le E_{balance,tolerance} \tag{12.4}$$

$$|SOE_{combined,final} - SOE_{combined,target}| \le \Delta SOE_{terminal} \tag{12.5}$$

The first gate rejects component sets that cannot satisfy requested wheel demand. The second detects inconsistent source, sink, or loss accounting. The third is applied to the Hybrid charge-sustaining comparison so a configuration cannot appear economical merely because it finishes with an excessive battery-energy deficit. A BEV mission is intentionally charge depleting, so the Hybrid terminal-SOE gate is inapplicable; BEV feasibility is governed by delivered mission energy, battery limits, and energy conservation. A rejection reason is stored when an applicable gate fails or a simulation raises an error.

12.6 Objective, ranking, and tie-break

$$\text{minimize } C_{km} = (C_{fuel} + C_{grid\text{-}equivalent\ electricity}) / d_{route} \tag{12.6}$$

Only feasible rows are sorted. Operating cost per kilometre is the primary objective; fuel consumption in litres per 100 km is the ascending tie-break. The returned MATLAB result contains the ranked feasible rows. Stable catalog IDs are retained so a result can be reproduced from the modular database.

12.7 What happens after the search

1. The calling script receives the number of evaluated combinations and the latest modeled cost through the progress callback when one is supplied.
2. All evaluated rows retain compatibility, feasibility, rejection reason, cost, energy, terminal SOE, unmet energy, and estimated-mass fields.
3. The returned TopConfigurations table contains the highest-ranked feasible rows, limited by the API result contract.
4. The best feasible result is returned to the caller; it does not mutate the Explorer app or its displayed simulation.
5. A cancellation callback can stop work between candidate simulations; completed rows remain available as a meaningful partial search history.

12.8 Worked ranking and responsible interpretation

The refreshed 144-case Battery 1 / Motor sweep retains the ten highest-ranked feasible rows. The leading row uses BAT-12 with MOT-12 at approximately 4.4411 EUR/km under the auxiliary-active-dump regeneration policy. Because all component values are synthetic, this ranking demonstrates workflow rather than a procurement recommendation.

Table 12.3. Top five feasible rows in the current 144-case study

Rank	Battery 1	Motor	Cost (EUR/km)	Electricity (kWh/km)
1	BAT-12	MOT-12	4.4411	1.6603
2	BAT-12	MOT-11	4.4425	1.6640
3	BAT-12	MOT-10	4.4431	1.6653
4	BAT-12	MOT-09	4.4436	1.6667
5	BAT-11	MOT-12	4.4531	1.6903

Table 12.4. Headless screening verification using the complete configuration handoff

API-shaped release audit	Evaluated	Feasible	Top sorted	Best returned
Hybrid default	40	4	Yes	Yes
BEV, 1.0 set / two packs	40	2	Yes	Yes; fuel = 0
BEV, 0.5 set / one pack	1 smoke case	1	Yes	Yes; Battery 2 count = 0

Correct interpretation
A first-ranked row is the best feasible row within the combinations that were actually evaluated, under the selected route, mass, prices, initial SOEs, component data, and terminal-energy rule. It is not automatically a production recommendation or a global optimum when the evaluation budget truncates the enabled search space.

13 Route Library and Provenance

13.1 Route categories

The database contains 20 routes: three official EU VECTO passenger missions, eight geographic long-distance coach corridors, and nine geographic German city cycles for Mannheim, Stuttgart, Berlin, Munich, Hamburg, Frankfurt, Cologne, Dusseldorf, and Leipzig. The former synthetic concept routes have been removed. Route_Catalog records region, source basis, organization, URL, retrieval time, licence, distance, duration, maximum speed, geolocation availability, endpoint coordinates, and notes.

13.2 Geographic route geometry and terrain

Route_Geometry stores 20,257 ordered latitude/longitude samples for seventeen geographic routes, together with cumulative polyline distance, geometry source, retrieval timestamp, cached elevation, and elevation provenance. Coordinates come from full GeoJSON route overviews returned by OSRM over OpenStreetMap road data. Terrain height is added as a separate Copernicus DEM GLO-90 channel for the 3D Route Map. The app can display either elevation in metres or derived slope in percent without removing the original 2D map. Elevation-derived slope is visualization evidence only: it is not silently substituted into the zero-grade long-route simulation channel.

Each German city circuit uses project-defined urban waypoints snapped to the driving network, a 50 km/h cap, 1.0 m/s2 acceleration and 1.3 m/s2 braking limits, and 20-second dwell events at nominal 800 m passenger-stop spacing. VECTO declaration cycles are not tied to a unique real road, so coordinates are intentionally not fabricated for them.

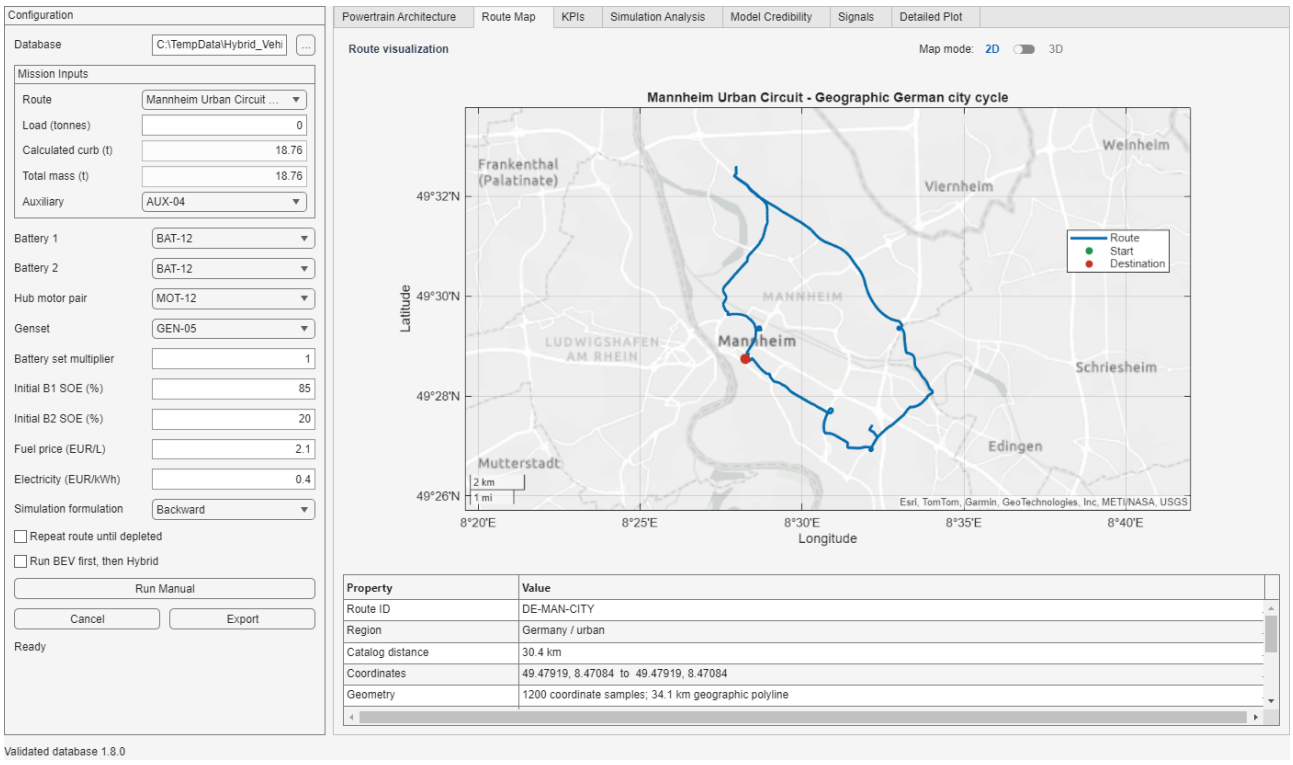


Figure 13.1. Mannheim geographic city circuit selected in the Route Map tab

13.3 VECTO missions

The Urban, Suburban, and Coach declaration-mode mission cycles are unmodified source files from the European Commission VECTO repository. The converter applies a deterministic bus driver with 1.0 m/s2 acceleration and 1.3 m/s2 braking bounds, honors declared stops, preserves the raw distance-domain trace, and produces a 1 Hz time history for this model.

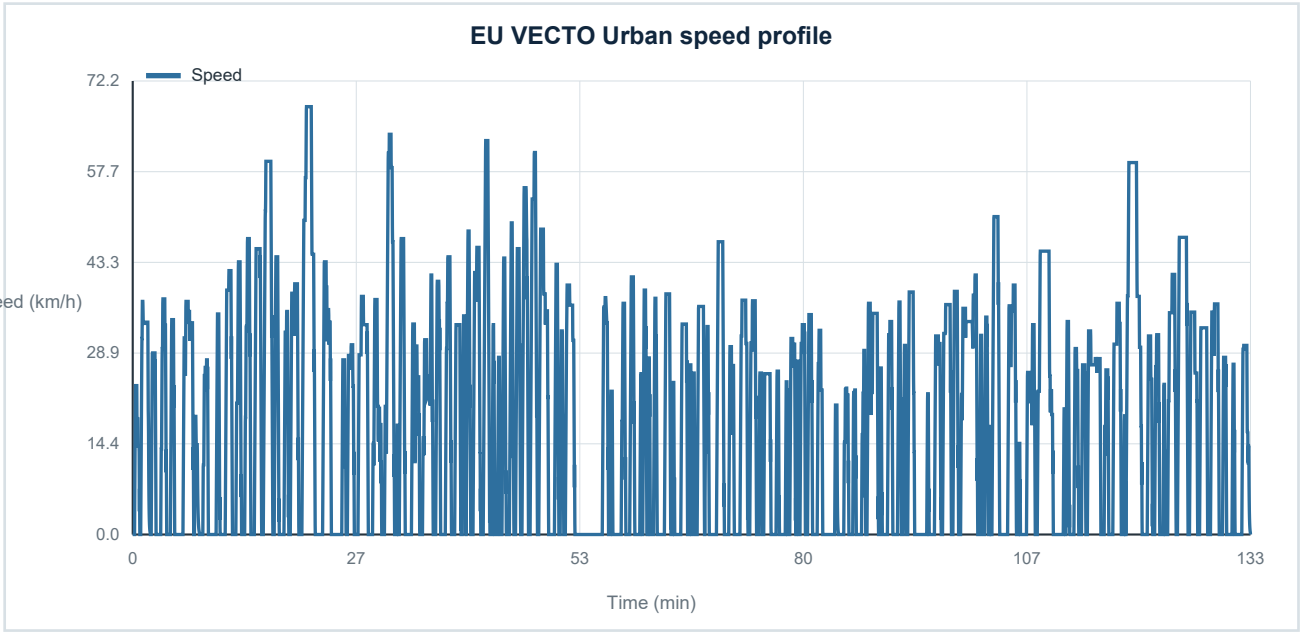


Figure 13.2. Default EU VECTO Urban mission after deterministic time conversion

13.4 Long European corridors

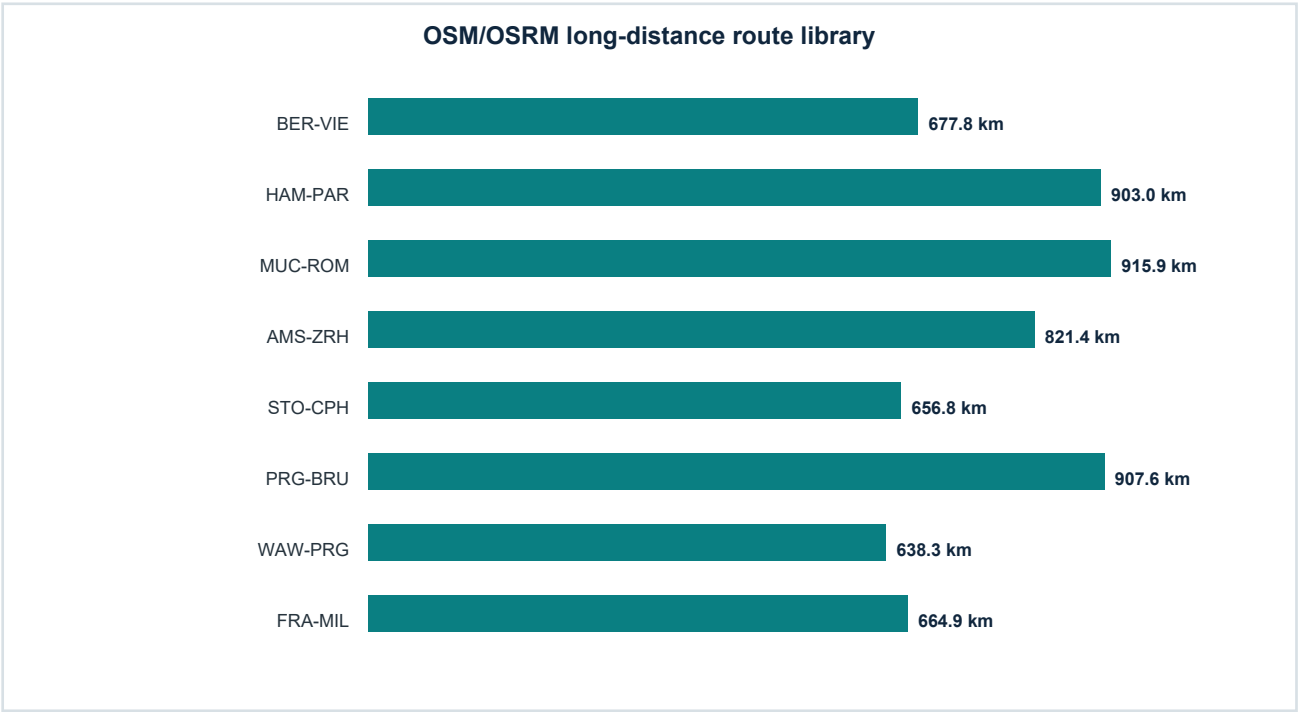


Figure 13.3. Geographic coach corridors in the requested 600-1000 km band

Table 13.1. Long-route source distances and adapted mission durations

Route	Countries	Distance	Adapted duration
Berlin, Germany to Vienna, Austria	Germany / Austria	677.8 km	8.45 h
Hamburg, Germany to Paris, France	Germany / France	903.0 km	11.18 h
Munich, Germany to Rome, Italy	Germany / Austria / Italy	915.9 km	11.54 h
Amsterdam, Netherlands to Zurich, Switzerland	Netherlands / Germany / Switzerland	821.4 km	9.73 h
Stockholm, Sweden to Copenhagen, Denmark	Sweden / Denmark	656.8 km	8.34 h
Prague, Czechia to Brussels, Belgium	Czechia / Germany / Belgium	907.6 km	11.12 h
Warsaw, Poland to Prague, Czechia	Poland / Czechia	638.3 km	8.32 h
Frankfurt, Germany to Milan, Italy	Germany / Switzerland / Italy	664.9 km	8.34 h

Route fidelity warning
OSRM supplies road-network distance and estimated segment durations but not elevation. Copernicus DEM GLO-90 now supports terrain visualization, but long-route simulation grade remains zero until a smoothed, road-aligned grade trace is derived and validated. Mountain-route energy is therefore still understated.

14 Verification, Validation, and Test Evidence

14.1 Layered evidence

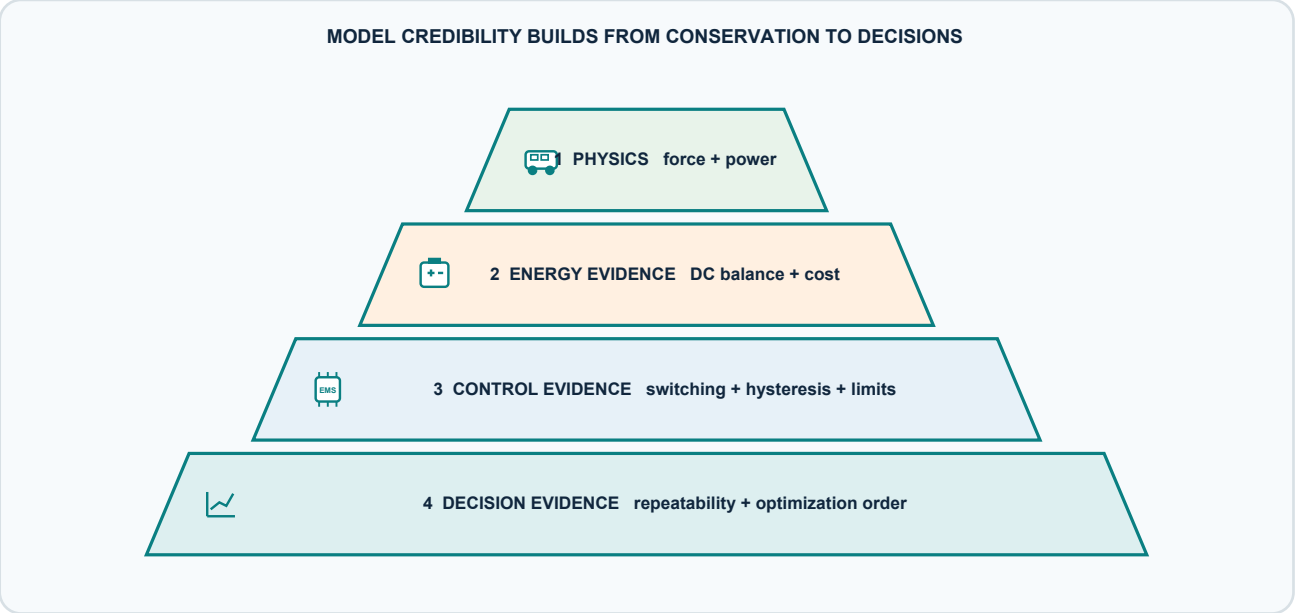


Figure 14.1. Verification pyramid used by the automated suite

The automated behavioral suite combines analytical comparisons, traction isolation, constant genset power, regeneration priority, pneumatic brake blending, modular data discovery, dynamic battery maps, motor loss maps, saturation, conservation, repeatability, optimizer ranking, scalable battery sets, calculated mass, constrained-speed behavior, depletion, and forward performance. All 50 behavioral scenarios currently pass. The complete MATLAB unit and integration suite contains 56 passing tests, including Hybrid/BEV implementation equivalence and component-extension coverage.

14.2 Test matrix

Table 14.1. Automated MATLAB test suite

Test	Purpose	Result
ConstantSpeedLevel	Constant-speed wheel power matches hand calculation	PASS
StationaryAuxiliary	Auxiliary load consumes battery energy at zero speed	PASS
LevelAcceleration	Acceleration increases wheel demand	PASS
PositiveGrade	Positive grade increases wheel power	PASS
DownhillRegeneration	Negative grade produces regenerative DC power	PASS
PneumaticBrakeBlending	Friction brakes supply braking demand beyond regeneration	PASS
RegenerationPriority	Regeneration supplies auxiliaries, then the active battery, then the load bank	PASS
BatterySwitching	Low active pack deterministically switches to standby pack	PASS
StandbyGensetCharge	Genset charges only the standby battery	PASS
GensetTractionIsolation	Genset output never offsets traction-bus demand	PASS
GensetHysteresis	Genset starts for low standby SOE without chattering	PASS
BatteryUpperLimit	Battery energy never exceeds upper limit	PASS
BatteryLowerLimit	Battery energy never falls below lower limit	PASS
MotorSaturation	Motor saturation reports unmet wheel power	PASS
GensetBestEfficiencyPoint	Genset runs at one constant optimum power	PASS
RouteGeolocation	Every geographic corridor has ordered valid latitude and longitude geometry	PASS
RouteMatStorage	Each route is loaded from one self-contained MAT file	PASS
ComponentMFileStorage	Each battery and motor is loaded from one versioned MATLAB data script	PASS
MotorLossMaps	Motor torque-speed maps are valid and determine conversion loss	PASS
BatteryDynamicMaps	Battery current limits, OCV, and resistance respond to SOE and temperature	PASS
GensetMFileStorage	Each complete genset assembly is loaded from one versioned MATLAB data script	PASS
ComponentExtension	Arbitrary battery, motor, and genset IDs load and simulate without central code changes	PASS
RouteValidationFailure	Duplicate route time is rejected	PASS
IncompatibleSelection	Compatibility filter records rejection reason	PASS
EnergyConservation	Cumulative DC residual remains within tolerance	PASS
Repeatability	Identical inputs produce identical scalar KPIs	PASS
ZeroSpeedStability	Zero-speed calculation remains finite	PASS
ComponentPhysicalSignals	Architecture component electrical and mechanical signals obey their defining equations	PASS
TerminalCostCorrection	Grid-equivalent terminal correction matches hand calculation	PASS
ChargeSustainingComparison	Terminal combined-SOE tolerance is enforced	PASS
OptimizationRanking	Feasible configurations are sorted by total cost per kilometre	PASS
ForwardFlatRoute	Forward plant completes a feasible flat route with achieved speed and distance	PASS
ForwardSevereGradeStall	An underpowered vehicle stalls without rollback and leaves the route incomplete	PASS

Test	Purpose	Result
ForwardDistanceIndexedGrade	Terrain changes are sampled at actual travelled distance	PASS
ForwardPowertrainModes	Forward formulation supports both BEV and Hybrid energy systems	PASS
FormulationDispatcher	The dispatcher preserves the legacy backward-demand result	PASS
ForwardSimulinkStructure	Editable forward Simulink plant compiles with the expected feature subsystems	PASS
ConstrainedAccelerationSuppression	Available current, torque, power, and force suppress achieved acceleration and speed	PASS
ConstrainedEnergyDepletion	A depleted battery leaves the vehicle stopped without rollback	PASS
ConstrainedDistanceTerrain	The fast constrained pass samples terrain at actual distance	PASS
ConstrainedPowertrainModes	The fast constrained pass supports both BEV and Hybrid	PASS
ConstrainedFixedHorizon	Constrained mode uses one route-time horizon and suppresses repeat-route expansion	PASS
BEVDualBattery	BEV uses both equal-SOE batteries in parallel with no fuel or genset	PASS
BEVSingleBattery	One-battery BEV electrically isolates Battery 2	PASS
HybridDoubleBatterySet	Two Hybrid sets scale both role banks equally	PASS
BEVThreeBatterySet	A 1.5-set BEV connects three packs with deterministic bank allocation	PASS
BatterySetValidation	Hybrid and BEV multipliers enforce their permitted increments	PASS
HybridCalculatedMass	Hybrid mass includes base curb, all batteries, genset, and load	PASS
BEVCalculatedMass	BEV mass includes base curb, all batteries, and load but no genset	PASS
BEVRegeneration	BEV regeneration follows auxiliary, batteries, resistor priority	PASS

14.3 Independent hand check

For the current calculated 18,763.636 kg default mass, the constant-speed level-road test compares model wheel power against an independent analytical expression. Both calculations give 20.542 kW at the test condition. The test also confirms that 15-tonne base curb, installed battery mass, genset mass, and entered load propagate into the longitudinal dynamics.

Passing tests do not validate synthetic data
Tests show that the implementation behaves consistently with its equations and stated limits. They do not prove that catalog efficiencies, BSFC values, component ratings, or route assumptions represent a production vehicle.

14.4 Requirements-to-evidence traceability

The release evidence package links each concept requirement to its equation or architectural rule, observable signal, verification scenario, and numerical acceptance criterion. This closes the argument from claim to evidence and makes missing proof visible. The complete machine-readable matrix is stored in Requirements_Traceability_Matrix.csv and the human-readable version is stored in documentation/Requirements_Traceability_Matrix.md.

Table 14.2. Current model-credibility gates

Credibility gate	Status	Decision
Requirements traceability	PASS	Release evidence is traceable
Physics and control verification	PASS	Concept behavior verified
MATLAB-Simulink equivalence	PASS	Independent Hybrid and BEV implementations agree within declared tolerances; do not infer measured-vehicle validation
Uncertainty screening	PASS	Use ranges for decision robustness, not statistical confidence
Concept baseline comparison	PASS	Suitable for concept screening, not procurement
Supplier-data calibration	NOT AVAILABLE	Obtain validated supplier maps
Measured-vehicle validation	NOT AVAILABLE	Collect vehicle measurements and define validation tolerances
Forward-facing feasibility	PASS	Use achieved-speed, gradeability, completion, and stall KPIs for concept screening
Fast constrained feasibility	PASS	Use for rapid screening; retain Performance mode for detailed mission closure
Thermal and ageing validation	NOT IMPLEMENTED	Add dynamic battery temperature and degradation states
Safety and fault analysis	NOT IMPLEMENTED	Perform FMEA/STPA and fault-injection verification

Gate discipline
A concept-verification PASS does not imply implementation-equivalence PASS, supplier calibration, or measured-vehicle validation. Management decisions must use the lowest unresolved gate relevant to the decision being made.

14.5 Independent implementation equivalence

The MATLAB kernels and Simulink models are independent implementations fed from the same selected configuration. Separate Hybrid and BEV cases compare speed, wheel demand, motor DC power, pneumatic friction braking, auxiliaries, both battery SOEs, genset power, total fuel, and integrated balance residual against declared tolerances. All twenty checks currently pass. This demonstrates numerical implementation equivalence for the declared cases; it is not supplier calibration or measured-vehicle validation.

Table 14.3. Hybrid and BEV MATLAB-Simulink signal-level equivalence

Mode	Signal	Max error	Tolerance	Unit	Status
Hybrid	Vehicle speed	0	0.02	m/s	PASS
Hybrid	Wheel demand	1.137e-13	5	kW	PASS
Hybrid	Motor DC power	0.1813	8	kW	PASS
Hybrid	Pneumatic friction brake	0	5	kW	PASS
Hybrid	Auxiliary power	0	0.05	kW	PASS
Hybrid	Battery 1 SOE	4.328e-07	0.025	fraction	PASS
Hybrid	Battery 2 SOE	2.22e-16	0.025	fraction	PASS
Hybrid	Genset power	0	1	kW	PASS
Hybrid	Total fuel	0.05455	0.1	L	PASS
Hybrid	Energy-balance integral	0	0.01	kWh	PASS
BEV	Vehicle speed	0	0.02	m/s	PASS
BEV	Wheel demand	1.137e-13	5	kW	PASS
BEV	Motor DC power	0.1729	8	kW	PASS
BEV	Pneumatic friction brake	0	5	kW	PASS
BEV	Auxiliary power	0	0.05	kW	PASS
BEV	Battery 1 SOE	2.064e-07	0.025	fraction	PASS
BEV	Battery 2 SOE	2.064e-07	0.025	fraction	PASS
BEV	Genset power	0	1	kW	PASS
BEV	Total fuel	0	0.1	L	PASS
BEV	Energy-balance integral	0	0.01	kWh	PASS

14.6 Decision robustness and concept baselines

The sensitivity study perturbs mass, aerodynamic drag, rolling resistance, auxiliary demand, motor efficiency, and battery discharge efficiency one factor at a time. It is deterministic screening, not a probability distribution. The concept comparison places the implemented hybrid beside analytical battery-electric and conventional-diesel screening baselines. Those baselines state their efficiency and fuel-energy assumptions and are suitable for architecture discussion, not procurement or homologation.

Table 14.4. Transparent powertrain concept screening

Concept	Cost (EUR/km)	Source energy (kWh/km)	Evidence
Proposed dual-battery hybrid	4.441	9.063	Implemented model
Battery-electric screening baseline	0.631	1.576	Analytical screening
Conventional-diesel screening baseline	1.253	5.845	Analytical screening

15 Worked Example: Default VECTO Urban Case

15.1 Case definition

The default Hybrid case uses the VECTO Urban route, one battery set containing two BAT-12 packs, two MOT-12 rear hub motors, GEN-05, 18.764 tonnes calculated curb mass, 0.0 tonnes entered load, and 18.764 tonnes total mass. Initial Battery 1/2 SOEs are 85% and 20%; energy prices are 2.10 EUR/L diesel and 0.40 EUR/kWh electricity.

15.2 Signal interpretation

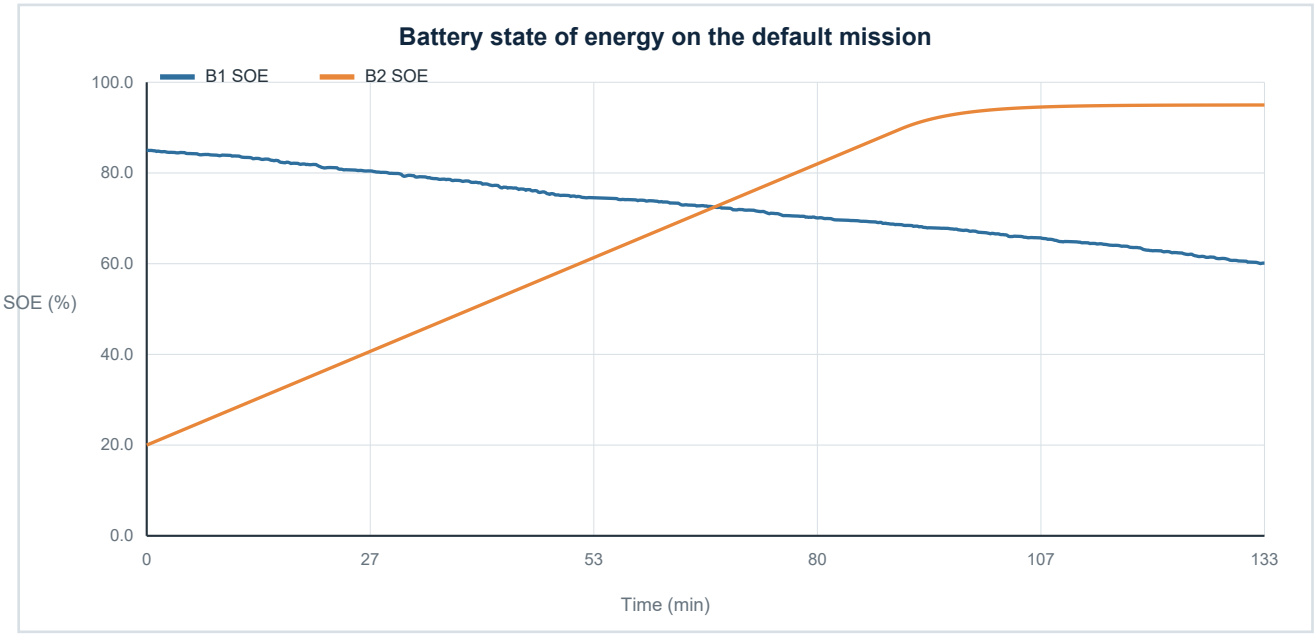


Figure 15.1. Default dual-battery SOE histories and active-pack behavior

Battery 1 is initially active and ends at 60.1% SOE; Battery 2 ends at 95.0% SOE. Regeneration is no longer redirected to the standby pack: 3.540 kWh supplies auxiliaries first, 23.689 kWh is accepted at the active-battery terminal, and 0.027 kWh is dissipated in the resistor bank. The genset does not start because the standby pack remains above the fixed 30% start condition.

15.3 Summary metrics

Table 15.1. Default-case results

Metric	Value	Interpretation
Route distance	39.787 km	distance integrated from the prepared 1 s trace
Grid-equivalent energy	66.060 kWh	terminal pack deficits divided by grid efficiency
Electrical intensity	1.6603 kWh/km	equivalent replenishment basis
Operating cost	4.4411 EUR/km	fuel plus grid electricity divided by distance
Available regenerated energy	27.256 kWh	non-negative energy available from negative motor DC power
Regeneration to auxiliaries	3.540 kWh	first-priority direct auxiliary supply
Regeneration to active battery	23.689 kWh	second-priority accepted terminal charge energy
Resistor-bank energy	0.027 kWh	third-priority controlled waste heat
Auxiliary energy	20.815 kWh	base and HVAC electrical demand
Unmet traction energy	0.000000 kWh	zero: case is feasible under the project criterion
Energy-balance error	0.000000000 kWh	zero within numerical reporting precision

Worked-example conclusion
The case is internally feasible and conservative under equivalent replenishment. It is not charge sustaining because combined terminal SOE is below the initial value, and it is not a real-vehicle performance claim.

16 Long-Route and Mass Feasibility Studies

16.1 Mass sweep

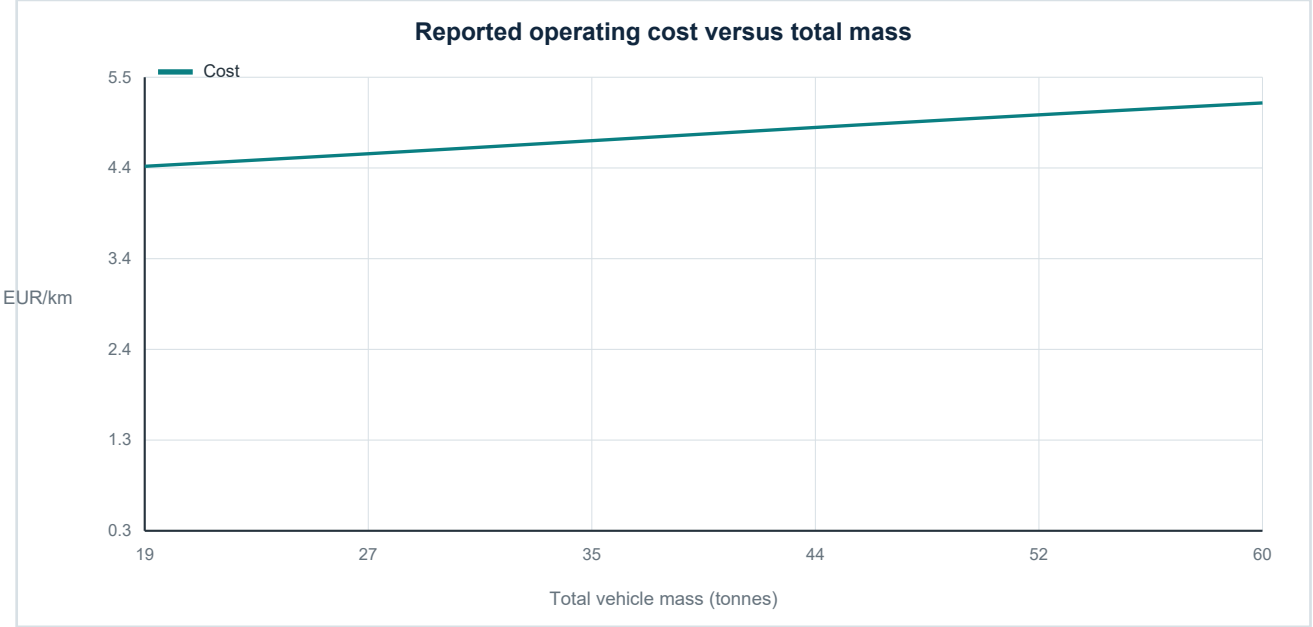


Figure 16.1. Cost trend must be read together with unmet traction energy

At 19 tonnes the default route has 0.000 kWh unmet traction energy. The entered payload is increased at each study point while the selected hardware-dependent curb mass remains fixed; unmet energy reaches 7.581 kWh at 60 tonnes. A cost trend alone therefore mixes feasible and infeasible cases; the feasibility flag is governing.

16.2 European long-route study

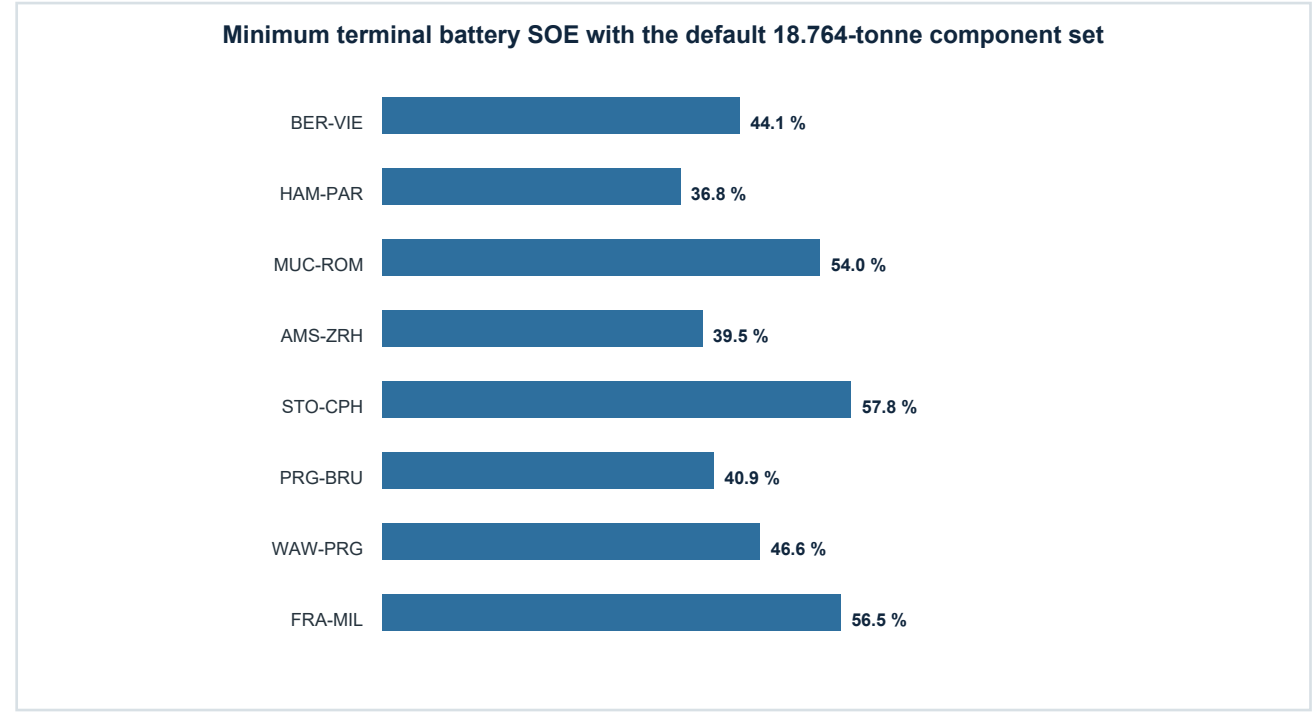


Figure 16.2. Minimum terminal pack SOE after one prescribed long-route pass

All 8 of 8 long-route cases serve the prescribed backward-demand trace without unmet traction/DC energy; unmet energy ranges from 0 to 0 kWh. This is an energy-accounting result, not proof of achieved-speed performance. Terminal battery reserve and fuel use vary materially, and the long-route dynamics grade remains zero until road-aligned elevation-derived grade is validated. These cases therefore support screening only and must not be presented as validated vehicle capability.

Table 16.1. Default-component long-route demand-service outcomes

Route	Simulated km	Fuel (L)	Unmet (kWh)	Demand served?
EUR-BER-VIE	677.4	272.3	0.0	Yes
EUR-HAM-PAR	902.2	360.1	0.0	Yes
EUR-MUC-ROM	915.0	371.9	0.0	Yes
EUR-AMS-ZRH	820.9	313.4	0.0	Yes
EUR-STO-CPH	656.3	268.9	0.0	Yes
EUR-PRG-BRU	906.8	358.2	0.0	Yes
EUR-WAW-PRG	638.1	268.0	0.0	Yes
EUR-FRA-MIL	664.8	268.8	0.0	Yes

16.3 How to make a long-route study meaningful

1. Increase motor and active-pack capability, add sufficient battery energy, or schedule external charging; do not count genset output as traction power.
2. Introduce realistic elevation and road-grade data before comparing Alpine corridors.
3. Calibrate speed, dwell, passenger load, HVAC, traffic, and rest-stop assumptions with measured or operator data.
4. Run the optional headless bounded screening API with feasibility constraints, then compare equivalent replenishment and charge-sustaining results.
5. Repeat under temperature, mass, headwind, and component-degradation scenarios.

17 Limitations, Responsible Use, and Extensions

17.1 Current model boundary

Table 17.1. Fidelity gaps and extension paths

Missing fidelity	Why it matters	Recommended extension
Battery dynamic temperature and ageing	Current maps use scenario temperature but do not integrate cell temperature or degradation	Calibrated electrothermal states plus cycle/calendar ageing
Ageing	Usable energy, resistance, and cost evolve with throughput and temperature	Cycle/calendar degradation and life-cycle cost
Driver and plant calibration	Forward and constrained formulations are implemented but use concept calibrations	Calibrate driver gains, force limits, stall thresholds, and plant response with vehicle data
Validated simulation grade for long routes	Cached terrain supports 3D display, but zero dynamics grade still understates mountain energy	Map-match, smooth, and validate an elevation-derived grade channel
Engine transients and emissions	Fuel and pollutants depend on warm-up and aftertreatment	Dynamic engine/aftertreatment model and measured maps
Wheel slip and lateral dynamics	Traction limits and stability are not represented	Tyre-road force limits and vehicle dynamics
Supplier data	Synthetic catalogs cannot support procurement	Version-controlled, validated manufacturer maps

17.2 Recommended development roadmap

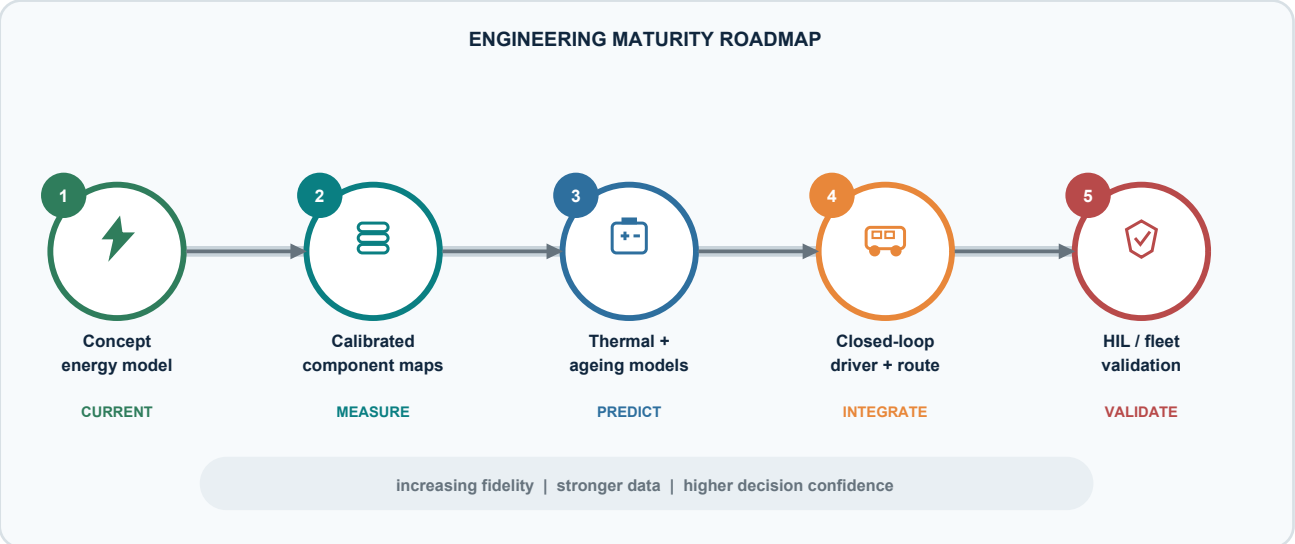


Figure 17.1. A disciplined progression from concept model to validated engineering platform

17.3 Graduate exercises

1. Derive an analytical constant-speed energy-per-kilometre expression and compare it with the kernel over several masses and headwinds.
2. Replace scalar motor efficiency with a two-dimensional torque-speed map and quantify the change in route energy.
3. Add elevation to one OSM/OSRM route, validate grade smoothing, and compare energy with the zero-grade baseline.
4. Design an alternative supervisor that allows controlled parallel discharge and define new safety/feasibility tests.
5. Compare 0.5-, 1.0-, 1.5-, and 2.0-set BEV operation, including capability sharing, route completion, terminal SOE, and mass.
6. Formulate a multi-objective search for cost, fuel, battery throughput, and component mass; produce a Pareto front.
7. Create a charge-sustaining urban case by sizing battery energy and the fixed-point standby charger; keep the 30% role threshold unchanged.
8. Add a simple battery thermal state and demonstrate a cold-weather power-derating scenario.
9. Construct a requirements-to-test traceability matrix for every limit and KPI in the model.

Final engineering lesson
A useful model is not only a set of equations. It is an agreement among data definitions, physical assumptions, interfaces, numerical methods, controller logic, evidence, and interpretation. When any one of those changes, the others must be reviewed.

Appendix A Quick-Start Laboratory

Use the following sequence for a first supervised laboratory session.

1. Open MATLAB R2025a and change to the project folder.
2. Run startup_hybrid_bus and confirm that the project loads without startup issues.
3. Run Results = run_hybrid_bus_simulation; inspect Results.Summary and Results.Validation.
4. Open HybridBus_BackwardModel.slx and trace signals from route input to energy accounting.
5. Launch app = HybridBusApp; repeat the default case and inspect all plot tabs.
6. Change only one factor, such as entered load or route, and explain every KPI change before running a second factor.
7. Run TestResults = run_all_hybrid_bus_tests; confirm 50 PASS behavioral scenarios, then run the full tests folder and confirm 56 passing unit/integration tests.
8. Run Credibility = generate_model_credibility_report; review every PASS, FAIL, NOT AVAILABLE, and NOT IMPLEMENTED gate before presenting a decision.

```
db = load_hybrid_bus_database(fullfile("data","HybridBus_ComponentDatabase.xlsx"));
report = validate_hybrid_bus_database(db);
assert(report.IsValid, strjoin(report.Errors,newline));

overrides = struct('SelectedRoute',"EUR-BER-VIE", ...
                   'PowertrainMode',"Hybrid", ...
                   'BatterySetMultiplier',1, ...
                   'LoadMass_t',5);
Results = run_hybrid_bus_simulation( ...
    fullfile("data","HybridBus_ComponentDatabase.xlsx"),overrides,'SaveResults',false);
Sequence = run_powertrain_sequence(db,overrides,true);
```

Appendix B Signal and Parameter Reference

Table B.1. Principal logged signals

Group	Signal	Unit	Meaning
Vehicle	Speed_m_s / Acceleration_m_s2	m/s, m/s2	prescribed speed and filtered acceleration
Wheel	Demand_kW / Delivered_kW	kW	required and limit-delivered wheel power
Wheel	UnmetTraction_kW / UnmetRegen_kW	kW	mechanical shortfalls
Motors	ElectricalPower_kW / MotorSpeed_rpm	kW, rpm	DC demand/regeneration and motor speed
Auxiliary	Power_kW	kW	base, HVAC, and route-multiplied load
Regeneration	Available_kW / ToAuxiliary_kW	kW	available braking energy and first-priority allocation
Regeneration	ToActiveBattery_kW / ResistorLoadBank_kW	kW	second- and third-priority allocations
Battery 1/2	Power_kW / Energy_kWh / SOE	kW, kWh, 1	pack source/sink and stored state
Genset	ElectricalPower_kW / FuelRate_L_s	kW, L/s	constant standby-charger power and fuel flow
Genset	ChargeDestinationBattery	integer	standby pack receiving genset power; zero when off
Controller	ActiveBattery / StandbyBattery / Mode	integer	supervisory role and discrete mode
Controller	ConnectedBatteryCount	integer	active-bank count in Hybrid; total connected pack count in BEV
Energy	BalanceResidual_kW	kW	power-conservation diagnostic
Energy	UnmetDCPower_kW / RejectedCharge_kW	kW	source/sink saturation accounting including load-bank power
Energy	RejectedGensetCharge_kW	kW	charger mismatch kept distinct from regenerative dissipation

Table B.2. Principal parameter groups

Area	Principal parameters	Workbook source
Vehicle	15 t base curb, installed battery/genset mass, entered load, gravity, Cd, frontal area	Calculated / Battery / Genset / Dashboard
Route	time, speed, grade, dwell, auxiliary multiplier	Route sheets
Hub drive	wheel radius, ratio, efficiencies, torque/power/speed	Tyre / Final Drive / Motor
Battery	usable energy, SOE bounds, efficiencies, power, regen, set multiplier, deterministic pack counts	Battery + Dashboard/app
Genset	optimum/max power, minimum times, BSFC, efficiency	Genset + maps
Supervisor	Hybrid 30% role swap or BEV parallel capability sharing; auxiliary-battery-dump regen priority	Architecture mode + control calibration
Load bank	regenerative surplus after active-pack limits	Derived power sink
Economics	fuel tank, prices, grid efficiency, terminal method	Dashboard / Vehicle / Prices / headless studies

References, Sources, and Project Files

[1] HybridBusProject source tree, modular route/component data, MATLAB functions, Hybrid, BEV, and Performance Simulink models, generated results, traceability matrix, and model-credibility evidence. Database version 1.8.0; textbook revision 2.7, revised 5 September 2026.

[2] European Commission, Vehicle Energy Consumption Calculation Tool (VECTO), https://climate.ec.europa.eu/eu-action/transport-decarbonisation/road-transport/vehicle-energy-consumption-calculation-tool-vecto_en.

[3] European Commission VECTO source repository, declaration mission cycles, commit d16ba783c1d8af0ea68797f5d7ed6cf01877d402, https://code.europa.eu/vecto/vecto.

[4] Project OSRM API documentation, route service and annotation objects, https://project-osrm.org/docs/v5.24.0/api/.

[5] OpenStreetMap contributors, Copyright and Open Database License attribution, https://www.openstreetmap.org/copyright.

[6] European Commission Mobility and Transport, Driving time and rest periods, including a 45-minute break after 4.5 hours of driving, https://transport.ec.europa.eu/transport-modes/road/social-provisions/driving-time-and-rest-periods_en.

[7] Project documentation: System_Architecture.md, Mode_Transition_Table.md, Parameter_Dictionary.md, Signal_Dictionary.md, Assumptions_and_Limitations.md, Test_Report.md, Model_Credibility_Report.md, Requirements_Traceability_Matrix.md, and Validation_Data_and_Extension_Plan.md.